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Semiconductors | North America

Disti Survey 3Q'26: Steady Progress, Tightening Supply

The recovery continues to make steady progress, with demand holding up well and supply conditions tightening.

AlphaWise α

Demand remains healthy across Analog/MCU, with above-seasonal expectations holding up and a constructive outlook heading into DecQ. While Analog growth expectations moderated from elevated levels, MCU strengthened, with both continuing to show solid demand relative to seasonality. Industrial remains a key driver of the recovery, with 55% of respondents reporting stronger-than-seasonal demand vs 48% last quarter, while Auto remains more mixed, with most respondents seeing demand broadly in-line with seasonal expectations.

Our cycle analysis points to further room for recovery, with volumes doing most of the work so far. Analog, MCU and IGBT/Power are all shipping above demand, though the recovery has been more gradual than in prior cycles. SIA unit trends have improved considerably, while ASPs remain below trend across the major product categories. With distributor inventory digestion winding down and supply constraints becoming more widespread, we see potential for pricing and inventory replenishment to contribute more meaningfully as the recovery progresses.

What does this mean for earnings? The continued cyclical recovery is encouraging for Analog/MCU suppliers, with demand improving and supply conditions gradually tightening. We're seeing more signs of widespread constraints, while secular opportunities such as AI power offer additional growth potential. The Industrial recovery is particularly constructive for ADI, TXN, and MCHP. Auto remains a slower recovery, though we still favor high-quality exposure via NXP.

Exhibit 1: 3Q26 Disti Survey Scorecard: Demand Holds Strong as Supply Tightens

3Q26 Disti Survey: Growth Expectations Remain Healthy; Supply Tightens			
Section	**Signal**	**Q/Q Trend | Reading**	
Growth Expectations	→ REMAIN HEALTHY	Analog ↓ MCU ↑ Power →	*Q/Q: 81st %ile | S: 87th %ile* *Q/Q: 98th %ile | S: 95th %ile*
End Markets	↗ BROADENING UPSIDE	AI/DC → Industrial ↑ Auto →	*95th %ile* *50th %ile*
Inventory/ Channel	→ STABLE, ELEVATED	Inventory → Double orders →	*77th %ile* *81st %ile*
Supply& Pricing	↗ TIGHTER SUPPLY, PRICING OFF HIGHS	Analog pricing ↓ MCU pricing ↓ Constraints ↑ Lead times ↑	*84th %ile* *84th %ile* *67th %ile* *64th %ile*

Source: AlphaWise, SIA, Morgan Stanley Research.

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Cycle Analysis: Steady Progress

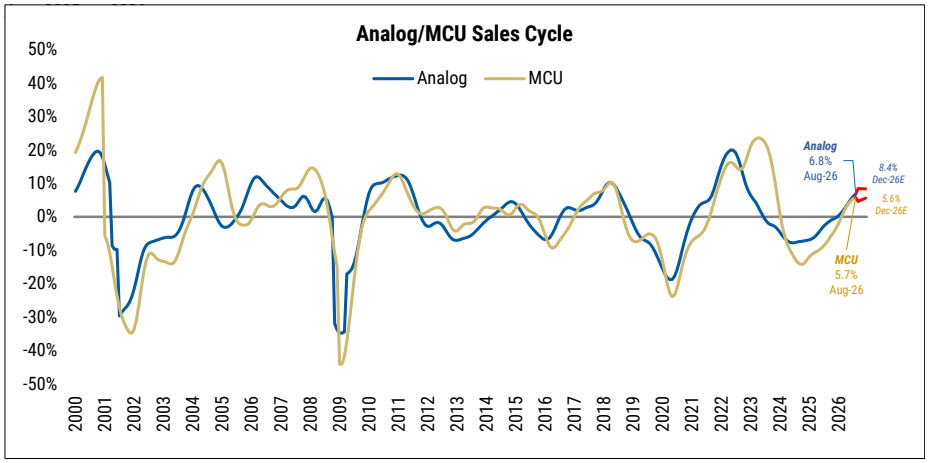
Analog, MCU and IGBT/Power are all shipping above demand, with Analog strengthening further and MCU Disti momentum accelerating in recent quarters. While the overall recovery remains gradual vs history, we see room for further acceleration as demand holds up, distributor inventories decline and pricing has yet to recover meaningfully.

Where Are We in the Cycle?

Analog, MCU and IGBT/Power are all shipping above demand, with the recovery continuing to broaden. Analog has strengthened further, while MCU demand momentum has picked up more recently, consistent with its later trough. The recovery has been slower than in prior cycles, but with inventories leaner and demand holding up, we think we're still in the early-to-middle stages of the cycle, rather than approaching a peak.

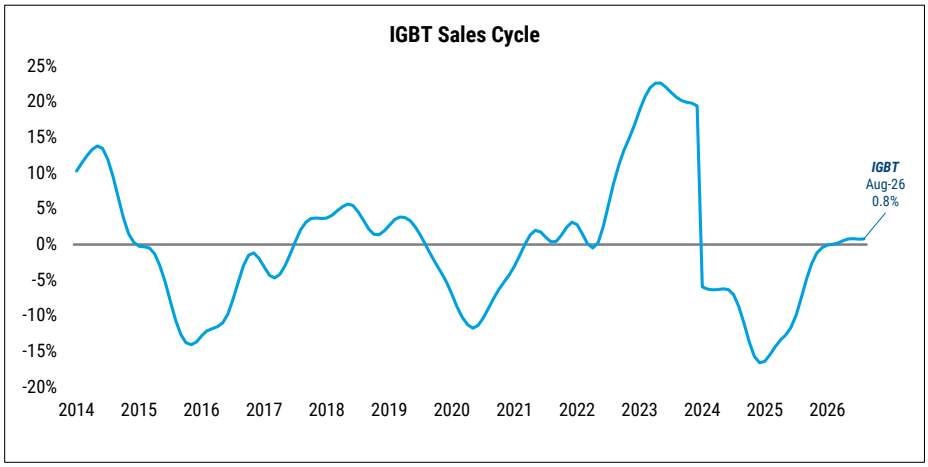
- **Analog:** Bottomed in June 2024 at 7.8% below demand. As of August 2026, Analog is shipping 6.8% above demand, a 14.6ppt improvement over 26 months. After a slow start, the recovery has gained momentum more recently.
- **MCU:** Bottomed later than Analog, in October 2024 at 14.2% below demand. As of August 2026, MCU is shipping 5.7% above demand, a 19.9ppt improvement over 22 months. The shipment indicator has continued to improve, while Disti momentum has accelerated in recent quarters, suggesting demand continues to strengthen.
- **IGBT/Power:** Moved from shipping ~1% below demand in February to 0.8% above demand in May, remaining at 0.8% in August. Power has also moved into modest overshipping, though the recovery has been more limited than in Analog and MCU.

Exhibit 2: Analog & MCU remain shipping above demand, both continuing to strengthen



Source: AlphaWise, SIA, Morgan Stanley Research.

Exhibit 3: IGBT remains modestly above demand, with limited improvement in recent months

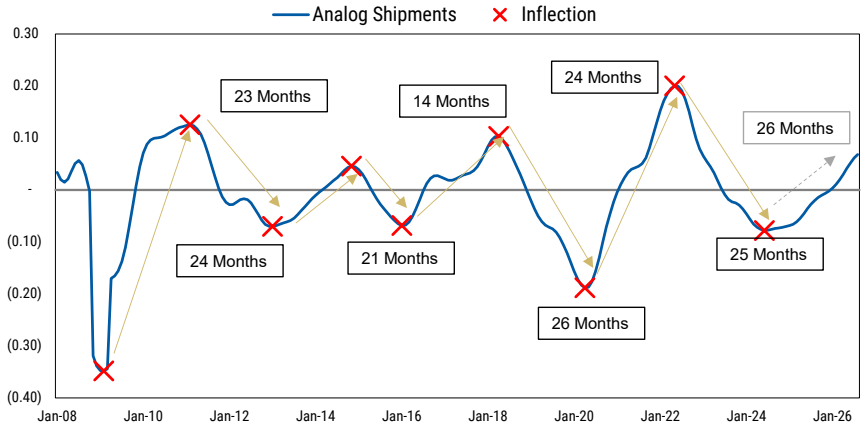


Source: AlphaWise, SIA, Morgan Stanley Research.

The recovery has been slower than in prior cycles, partly because semiconductor suppliers have been carrying elevated inventories, allowing them to meet improving demand without a broad customer restocking cycle. Distributor inventories have continued to decline, while customers remain cautious about rebuilding stock. We expect the recovery to remain gradual near term, though rising lead times and broader supply constraints could eventually push customers to replenish inventories more aggressively, providing further upside.

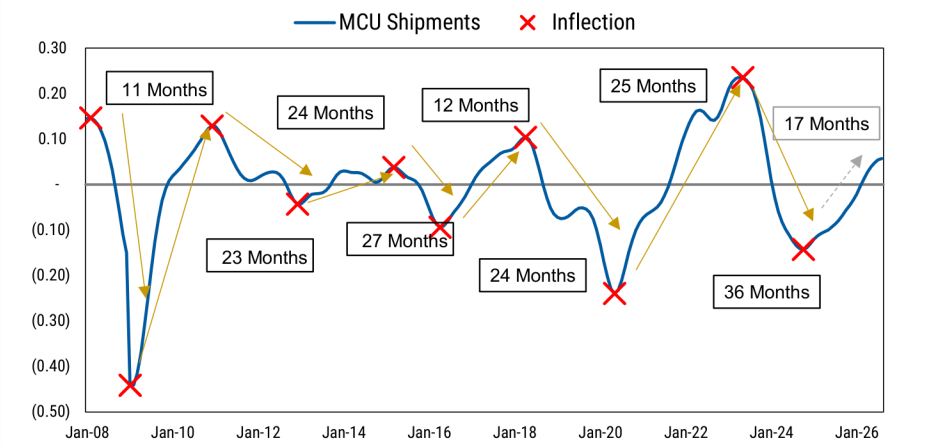
The length of the recovery doesn't necessarily mean we're approaching a peak. Analog is now 26 months past its June 2024 trough, at the upper end of the ~22-26 months historically seen between cycle inflection points, while MCU bottomed four months later. Still, we don't see clear signs of a peak, given the slow initial recovery and continued improvement in underlying demand.

Exhibit 4: Analog cycles have typically lasted ~22-26 months between cycle peaks & troughs



Source: AlphaWise, SIA, Morgan Stanley Research.

Exhibit 5: MCU cycle duration has been more variable, with pre-2020 cycle legs typically lasting ~22-27 months



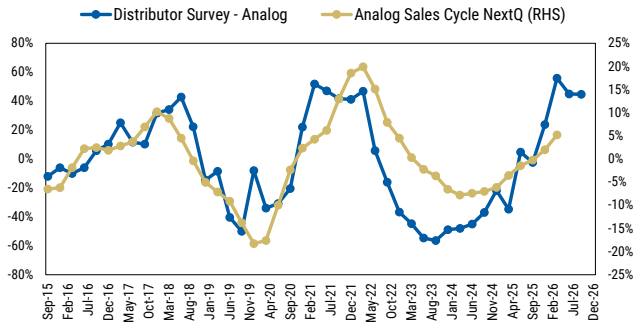
Source: AlphaWise, SIA, Morgan Stanley Research.

Disti Momentum - Staying Strong as Inventory Digestion Winds Down

Disti momentum remains strong, with Analog holding at elevated levels and MCU accelerating again in 3Q. The tracker measures the spread between respondents building and depleting inventory, and the latest strength reflects a meaningful decline in inventory digestion rather than a broad increase in restocking. Across the survey, respondents planning to deplete inventory fell to 13% from 30% last quarter, while those planning to build remained relatively steady at 47%. For Analog, momentum has held up following the sharp improvement earlier this year, while MCU has gained further traction more recently. We see this as a constructive sign that the channel is moving past the inventory correction, with room for another leg of improvement if tightening supply prompts customers to rebuild stock more aggressively.

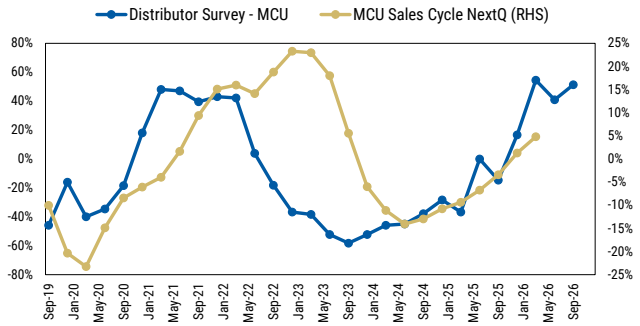
The return implications are somewhat more nuanced. Historically, forward returns for Analog and MCU have generally tended to be weaker as shipments move further above demand, although there has been considerable variation across cycles. This suggests that stock selection may become more important as the cycle progresses. However, with the recovery still gradual and broad restocking yet to materialize, we don't view the historical relationship as a reason to turn negative on the group.

Exhibit 6: Analog Disti momentum remains high following the sharp improvement earlier this year...



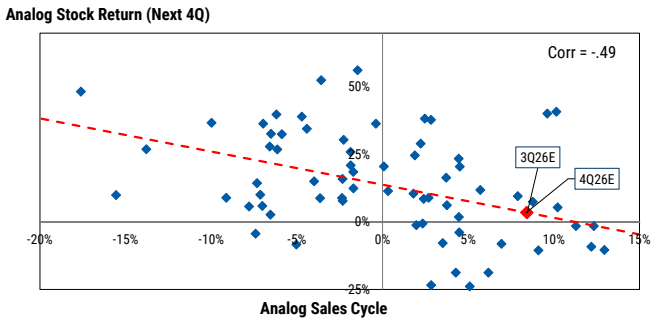
Source: AlphaWise, SIA, Morgan Stanley Research

Exhibit 7: ...while MCU Disti momentum has accelerated again, pointing to further improvement in the inventory cycle.



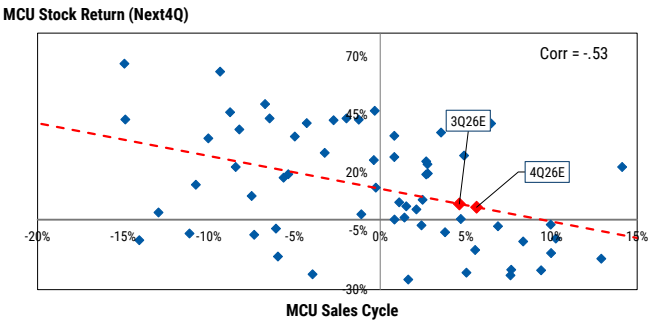
Source: AlphaWise, SIA, Morgan Stanley Research

Exhibit 8: Forward analog returns historically moderate as the sales cycle advances, but there has been variation...



Source: AlphaWise, SIA, Morgan Stanley Research

Exhibit 9: ...with MCU forward returns showing a similar historical tendency



Source: AlphaWise, SIA, Morgan Stanley Research

Disti Survey 3Q'26: Healthy Demand as Supply Tightness Broadens

The recovery has continued to make steady progress following the sharp 1Q acceleration, which marked the beginning of a broader Industrial recovery. Demand remains healthy in SepQ, with constructive expectations heading into DecQ, while Auto remains mixed. Pricing remains firm, inventory digestion is winding down and supply constraints are becoming more widespread. With volumes leading the recovery so far, we see further potential upside from pricing and inventory replenishment.

AlphaWise Analysis: Healthy Demand, Tightening Supply

Our 3Q26 Distributor Survey points to continued strength in Analog/MCU demand, with a healthy SepQ and constructive expectations heading into DecQ. For SepQ, 61% of respondents expected sequential Analog growth, down from 68% last quarter, while MCU expectations improved to 62% from 51%. Looking ahead to DecQ, 63% expect Analog growth and 65% expect MCU growth, with MCU expectations improving 11ppt q/q. Demand also remains healthy relative to seasonal trends, with 95% of respondents expecting Analog and 100% expecting MCU to perform in-line with or above seasonal levels. Industrial remains a bright spot, with net positive responses increasing to 53% from 40%, while Auto continues to show a more mixed recovery.

Supply conditions are also tightening as demand improves. The share of respondents reporting supply constraints rose to 73% from 68%, while those seeing longer lead times increased to 63% from 53%. Inventory-building intentions remained relatively steady at 47%, but the share depleting inventory fell to 13% from 30%, suggesting the inventory correction is winding down. Pricing remains firm, though fewer respondents expect further increases across Analog and MCU. Taken together, the survey points to a healthier demand environment and tightening availability, but not yet the broad inventory rebuild that could drive a more pronounced acceleration in the cycle.

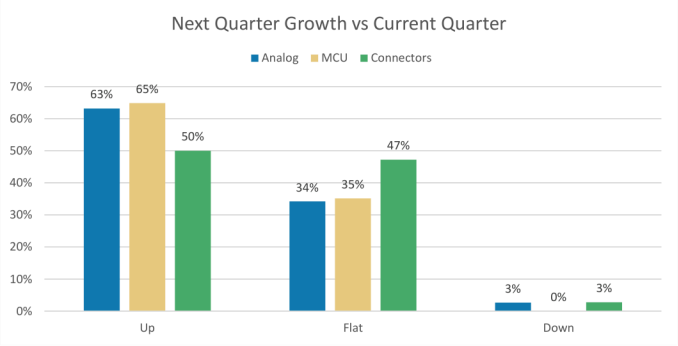
Exhibit 10: 3Q26 Disti Survey Scorecard:

3Q26 Disti Survey: Growth Expectations Remain Healthy; Supply Tightens

Section	Signal	Q/Q Trend | Current Reading	Survey Quotes
Growth Expectations	→ REMAIN HEALTHY	Analog ↓ Analog growth: 63% q/q (-12pp); 81st %ile MCU ↑ MCU growth: 65% q/q (+11pp); 98th %ile Power → Power demand: 66% higher (+3pp)	Analog: MCU: "... across all markets especially Industrial" Power: "All power management devices, isolation, all end markets"
End Markets	↗ BROADENING UPSIDE	AI/DC → 66% higher demand (+3pp) Industrial ↑ 53% net positive (+13pp); 95th %ile Auto → 16% net positive (+3pp); 50th %ile	AI/DC: "Yes high power ic for data center." Industrial: "...most customers are buying ahead in industrial, medical and defense/aero mkts." Auto: "...industrial automation, automotive, and many traditional electronics segments remain mixed..."
Inventory/ Channel	→ STABLE, ELEVATED	Inventory → 47% build (-3pp); 77th %ile Double orders → 34% (-3pp); 81st %ile	Restocking: "...any long lead time component that is single source." Orders: "Double ordering appears to be limited and targeted rather than widespread across the market." Urgency: "...all markets have been Expediting product across all commodities."
Supply& Pricing	↗ TIGHTER SUPPLY, PRICING OFF HIGHS	Analog pricing ↓ 55% stronger (-7pp); 84th %ile MCU pricing ↓ 54% stronger (-5pp); 84th %ile Constraints ↑ 73% constrained (+5pp); 67th %ile Lead times ↑ 63% rising (+11pp); 64th %ile	Analog: "lead-time and demand driving pricing increases" MCU: "lead time going out, demand pressure driving pricing increases" Power: "Infineon power parts Lead Times pushing out"

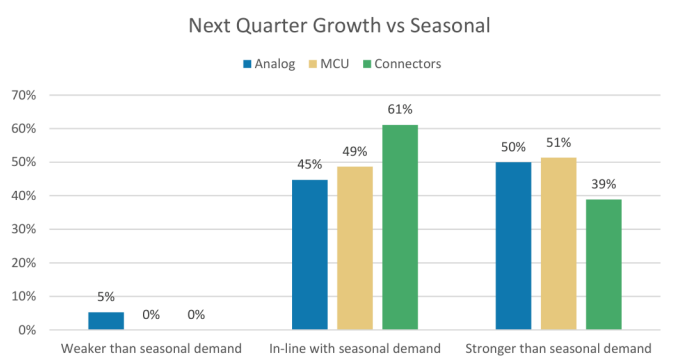
Source: AlphaWise, SIA, Morgan Stanley Research.

Exhibit 11: Disti results point to positive sequential growth for the SepQ...



Source: AlphaWise, SIA, Morgan Stanley Research

Exhibit 12: ...while seasonally adjusted expectations remain healthy across Analog/MCU.



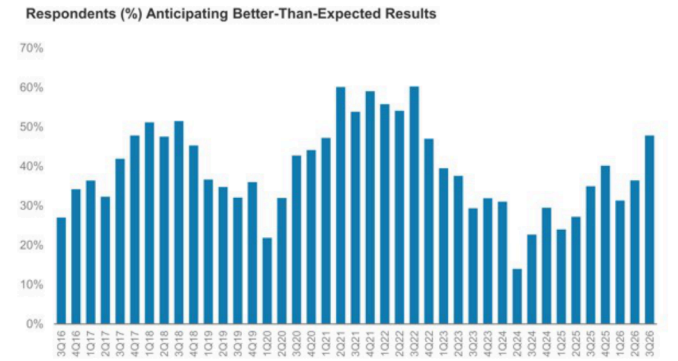
Source: AlphaWise, SIA, Morgan Stanley Research

1) Industrial Demand Strengthens; Auto Still Mixed

Industrial continues to be a bright spot, with net positive responses rising to 53% from 40%, putting the latest reading in the 95th percentile of historical results. We're seeing a broader recovery take hold across Industrial, rather than strength being concentrated in just a few end markets. Our latest Multi-Industry Distributor Survey also points to continued growth, though FY26 volume expectations eased slightly, largely reflecting seasonality.

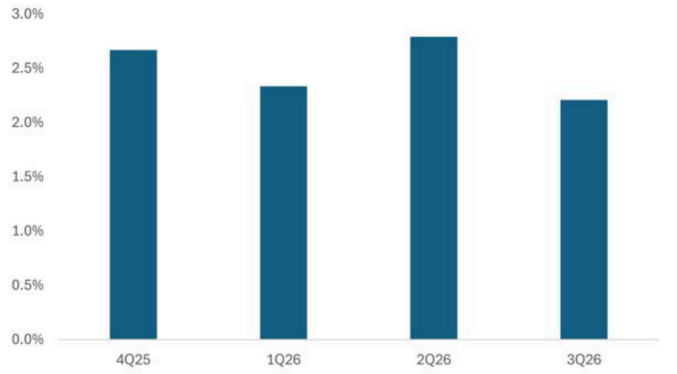
Auto has been slower to recover. Net positive responses improved modestly to 16% from 13%, but the underlying trends remain mixed. Inventory digestion appears to be largely complete, with suppliers shipping closer to underlying demand, though we're not yet seeing the same breadth of improvement as in Industrial.

Exhibit 13: Multi-Industry distributors continued to anticipate better-than-expected results in 3Q26...



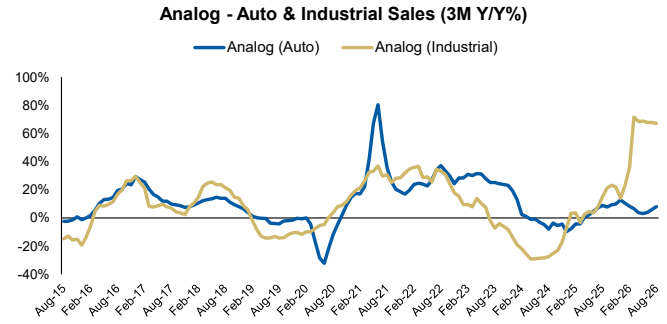
Source: AlphaWise, Morgan Stanley Research

Exhibit 14: ...and while FY26 volume expectations moderated slightly, our Multi-Industry colleagues see this as largely seasonal



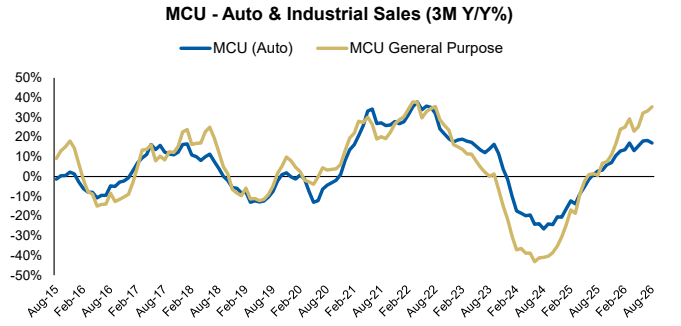
Source: AlphaWise, Morgan Stanley Research

Exhibit 15: Industrial analog sales snapped back sharply, outpacing Automotive...



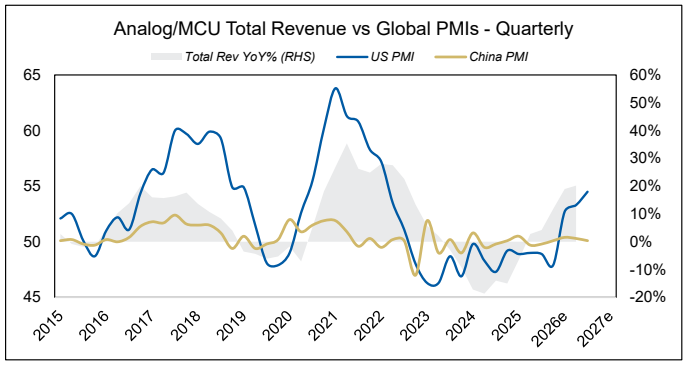
Source: SIA, Morgan Stanley Research.

Exhibit 16: ...with MCU showing a similar trend



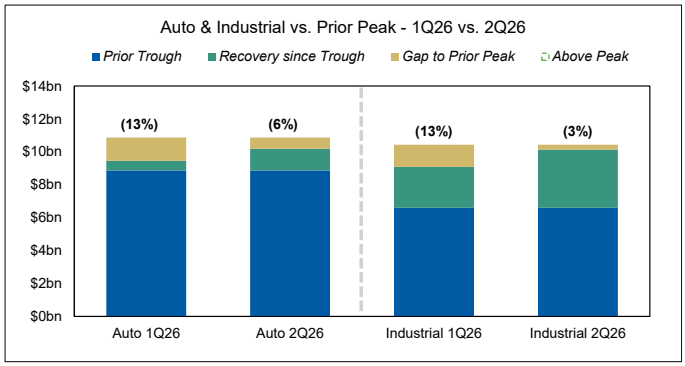
Source: SIA, Morgan Stanley Research.

Exhibit 17: PMIs have been in expansionary territory since January of this year...



Source: Company data, VisibleAlpha, Morgan Stanley Research

Exhibit 18: ...supporting Industrial recovering faster than auto, though both remain below prior peaks



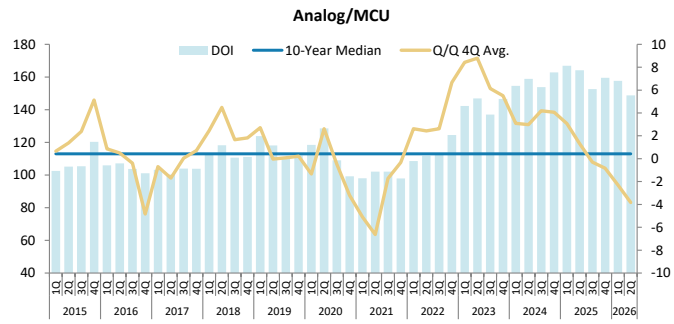
Note: As of 2Q26 earnings actuals. Source: Company data, VisibleAlpha, Morgan Stanley Research.

2.) Inventory Depletion Eases Sharply; Builds Remain Healthy

Inventory digestion continues to wind down, with only 13% of respondents planning to deplete inventory vs 30% last quarter. Meanwhile, the share planning to build inventory remained relatively steady at 47%. The improvement is also showing up across products: Analog depletion fell to 5% from 13%, while MCU inventory-building intentions increased to 46% from 38%.

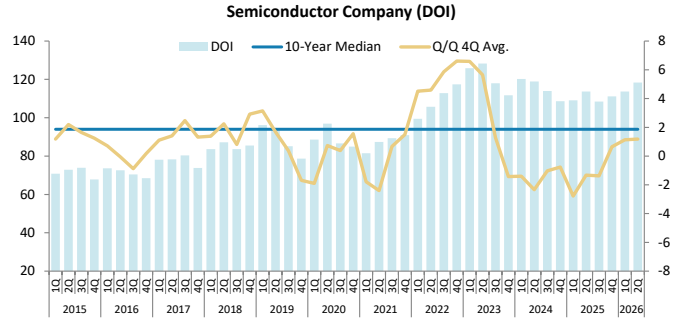
Distributors are increasingly moving away from inventory reductions, but they're not yet rebuilding stock aggressively. With supplier inventories still elevated, the recovery can continue without a major restocking cycle, leaving that as a potential additional driver further down the road.

Exhibit 19: Analog/MCU inventory remains above historical median, but DOI momentum has continued to moderate since 3Q23...



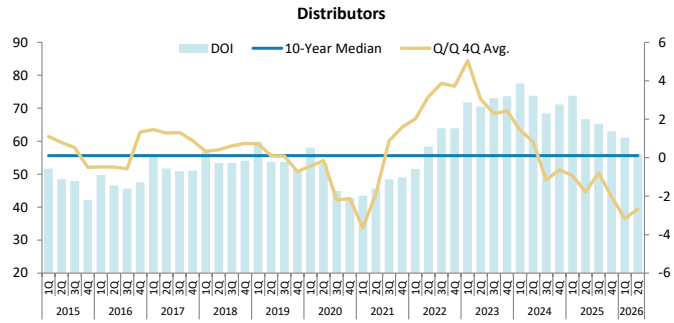
Source: FactSet, Morgan Stanley Research

Exhibit 20: ...while overall semi company DOI momentum has been increasing...



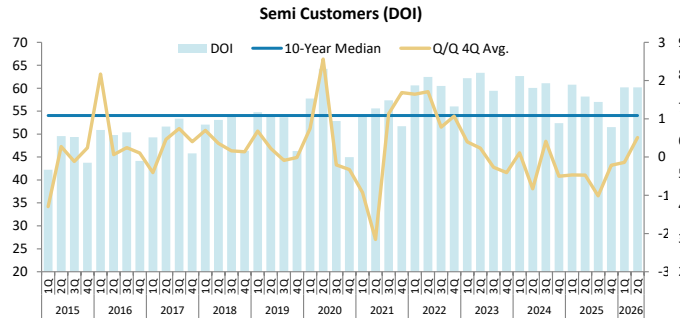
Source: FactSet, Morgan Stanley Research

Exhibit 21: ...though distributor inventory has declined for 6 consecutive quarters...



Source: FactSet, Morgan Stanley Research

Exhibit 22: ...while customer DOI remained flat q/q, restocking ~in-line with 2Q seasonality.

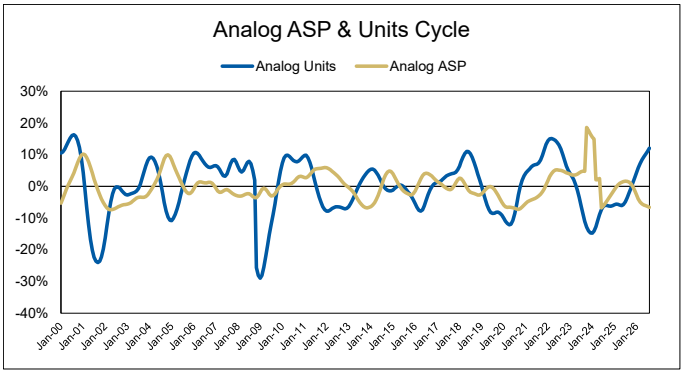


Source: FactSet, Morgan Stanley Research

3) Pricing Remains Firm as Supply Tightness Broadens

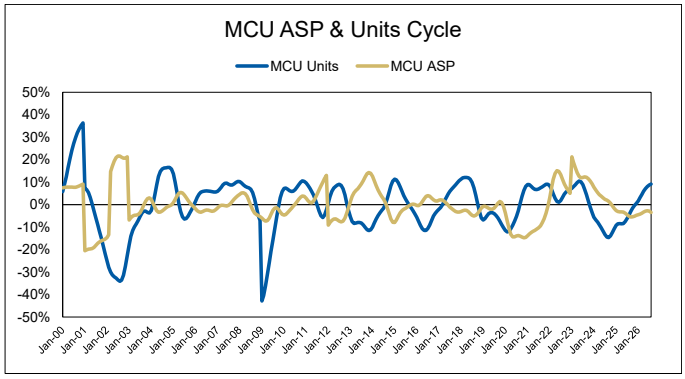
The SIA data continue to point to a recovery driven primarily by volumes, while pricing has been slower to follow. Analog units are now ~12% above trend, up from ~4% below a year ago, while Power Analog units have reached a record ~22% above trend. MCU units have also recovered to ~9% above trend, though IGBT momentum has moderated, with units ~5% above trend vs ~10% in May. ASPs remain below trend across all four categories, with Analog and Power Analog particularly weak at ~7% and ~9% below trend, respectively. MCU and IGBT pricing has improved from prior troughs, but a broad ASP recovery has yet to take hold. Meanwhile, the latest distributor survey points to further tightening in supply, with 73% reporting constraints vs 68% last quarter, alongside rising lead times and expedite activity. With volumes leading the recovery so far, we see potential for pricing to contribute more meaningfully as demand continues to improve and supply availability tightens.

Exhibit 23: Analog units have recovered to ~12% above trend, while ASPs remain ~7% below trend...



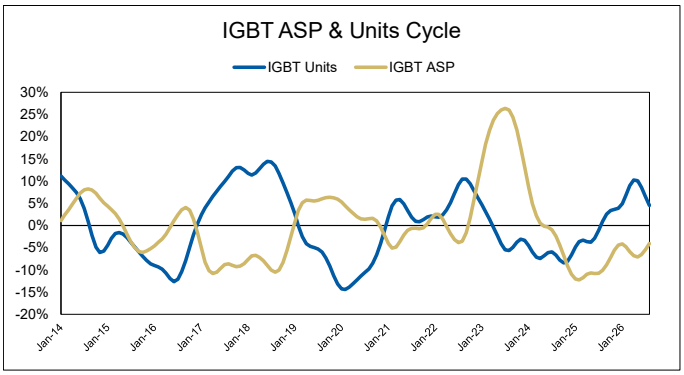
Source: AlphaWise, SIA, Morgan Stanley Research

Exhibit 24: ...and MCU units are ~9% above trend, though ASPs remain below trend despite recovering from their trough



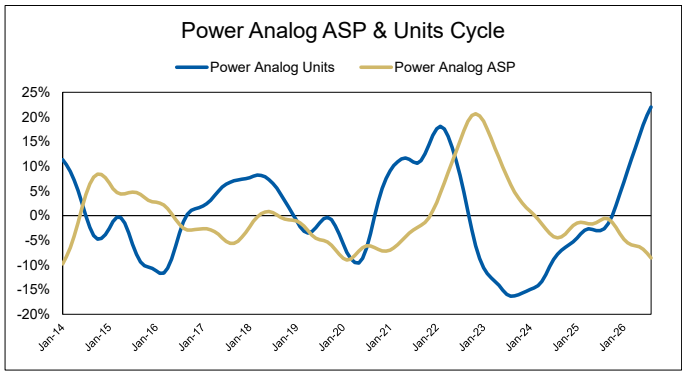
Source: AlphaWise, SIA, Morgan Stanley Research

Exhibit 25: IGBT units remain above trend despite recent moderation, while ASPs continue to recover from their trough



Source: AlphaWise, SIA, Morgan Stanley Research

Exhibit 26: Analog Power Mgmt units have reached record levels above trend, while ASPs remain near historical lows

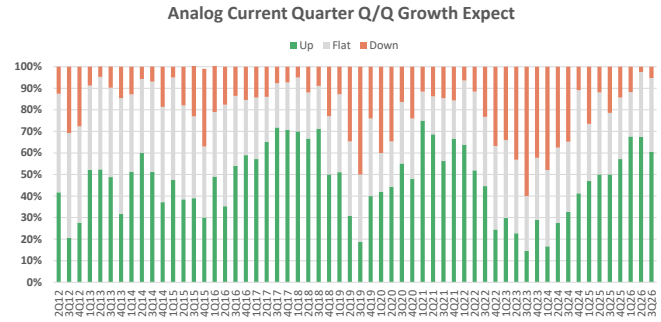


Source: AlphaWise, SIA, Morgan Stanley Research

3Q26 Results - Key Charts & Takeaways

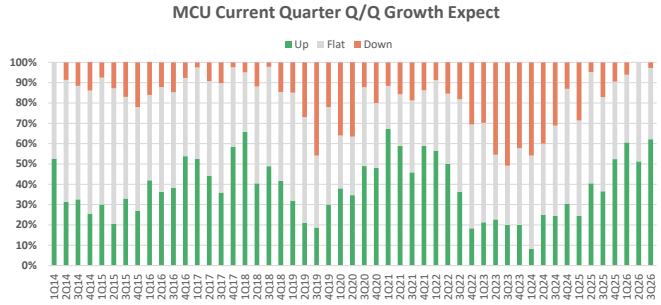
Main takeaway: Another strong quarter for SepQ, with healthy DecQ expectations as supply tightens

Exhibit 27: Analog sequential growth expectations for SepQ moderated to 61% from 68%, but remain healthy...



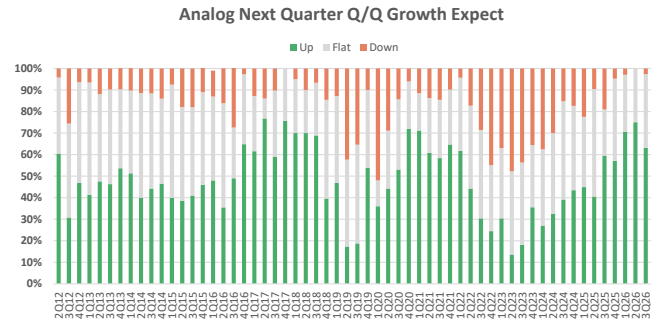
Source: AlphaWise, Morgan Stanley Research

Exhibit 28: ...while MCU expectations strengthened to 62% from 51%, with both product lines showing solid growth expectations



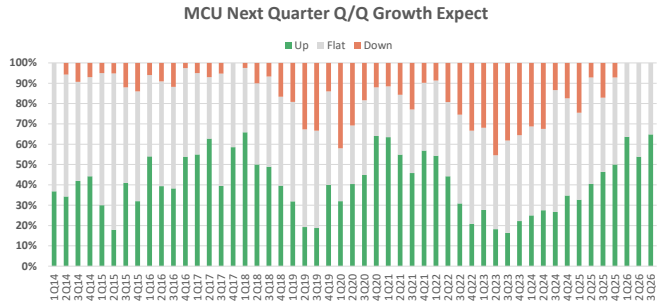
Source: AlphaWise, Morgan Stanley Research

Exhibit 29: Analog growth expectations for DecQ moderated to 63% from 75%, though the majority still anticipate sequential growth...



Source: AlphaWise, Morgan Stanley Research

Exhibit 30: ...while MCU expectations strengthened to 65% from 54%, with no respondents anticipating a decline

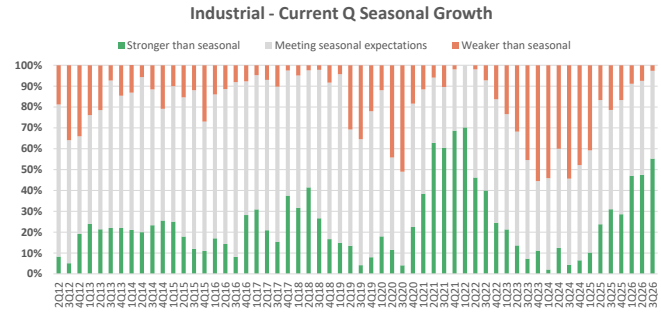


Source: AlphaWise, Morgan Stanley Research

1) Industrial Accelerates Again, Auto Still Mixed

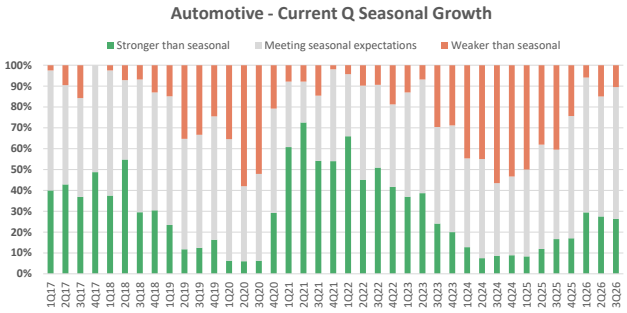
Industrial continues to lead the broader recovery, with stronger-than-seasonal expectations increasing to 55% from 48% last quarter. The improvement was also evident in programmable logic, where above-seasonal responses rose to 41% from 29%. Auto remains less convincing, although weaker-than-seasonal responses declined to 11% from 15%, pointing to some stabilization. Taken together, the data suggest that the Industrial recovery is gaining traction, while Auto continues to lag.

Exhibit 31: Stronger-than-seasonal Industrial expectations increased q/q



Source: AlphaWise, Morgan Stanley Research

Exhibit 32: ...while weaker-than-seasonal Auto expectations decreased to 11% from 15%

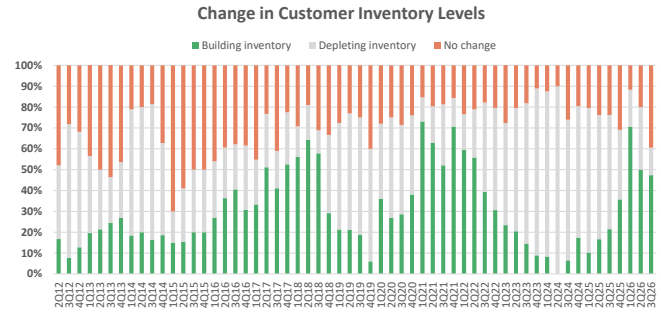


Source: AlphaWise, Morgan Stanley Research

2) Inventory Depletion Eases Sharply; Builds Remain Healthy

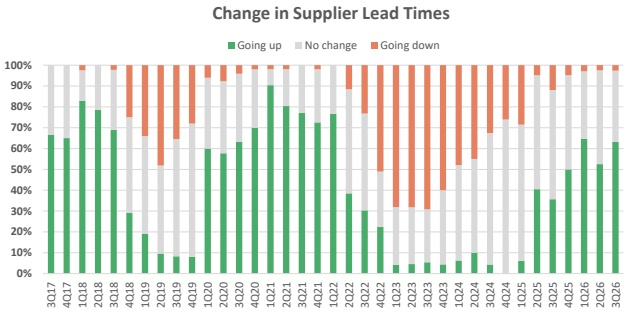
Inventory trends improved in 3Q, with depletion intentions falling to 13% from 30%, while the share of respondents planning to build inventory remained relatively steady at 47%. At the product level, Analog depletion fell to 5% from 13%, while MCU inventory-building intentions increased to 46% from 38%. Builds greater than 5% also increased modestly across Analog/MCU. The decline in depletion suggests that inventory digestion is easing, although the relatively stable overall building intentions do not yet point to broad-based restocking.

Exhibit 33: Those looking to deplete inventory decreased from 30% in the JunQ to 13% in the SepQ...



Source: AlphaWise, Morgan Stanley Research

Exhibit 34: ...while respondents reporting rising supplier lead times increased to 63% from 53%...

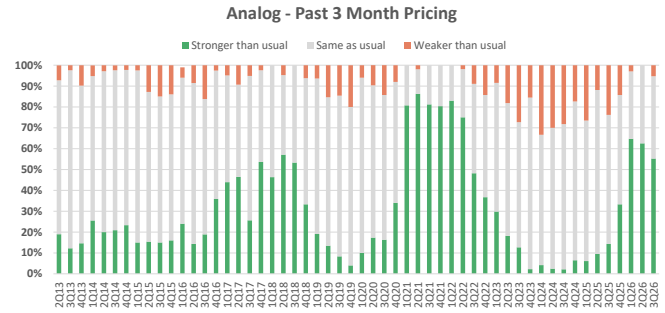


Source: AlphaWise, Morgan Stanley Research

3) Pricing Moderates as Supply Tightness Broadens

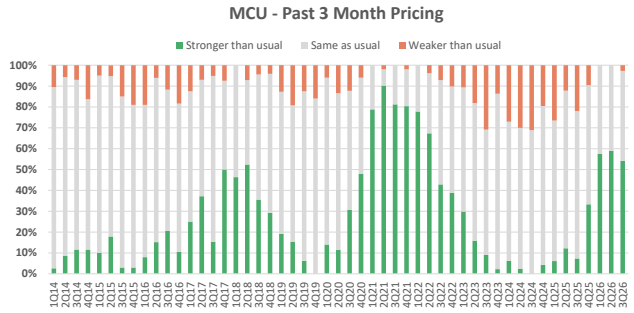
Pricing remains firm despite some moderation from 2Q highs. Respondents reporting stronger-than-usual pricing declined to 55% from 63% for Analog and 54% from 59% for MCU, although both readings remain elevated historically. Supply conditions, meanwhile, tightened further, with overall constraints increasing to 73% from 68% and respondents reporting rising lead times climbing to 63% from 53%. Expedite activity also increased to 68% from 58%, while double ordering moderated to 34% from 38%. We see broader evidence of supply tightness, though pricing trends suggest that this has not yet translated into another leg of accelerating pricing pressure.

Exhibit 35: Stronger-than-usual Analog pricing decreased from 63% to 55%...



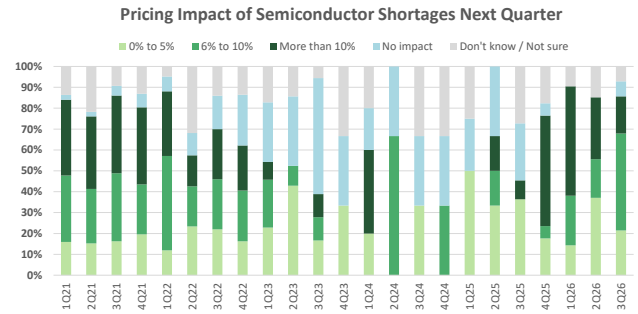
Source: AlphaWise, Morgan Stanley Research

Exhibit 36: ...while 54% of respondents saw stronger MCU pricing, down from 59% last quarter...



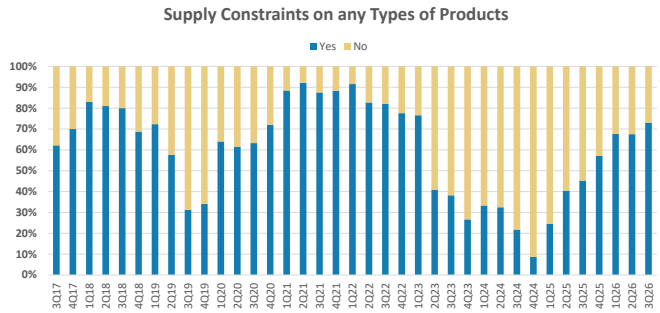
Source: AlphaWise, Morgan Stanley Research

Exhibit 37: ...and the magnitude of pricing pressure remained elevated...



Source: AlphaWise, Morgan Stanley Research

Exhibit 38: ...while overall supply constraints increased to 73% from 68%



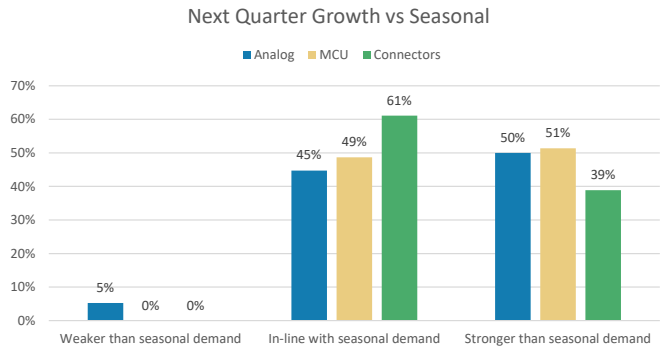
Source: AlphaWise, Morgan Stanley Research

3Q'26 Disti Survey - Details

In conjunction with Morgan Stanley AlphaWise, we conducted our 55th quarterly survey among 38 semiconductor distributors during August - October 2026. We asked questions around sales volumes, growth expectations and inventory levels, among other topics.

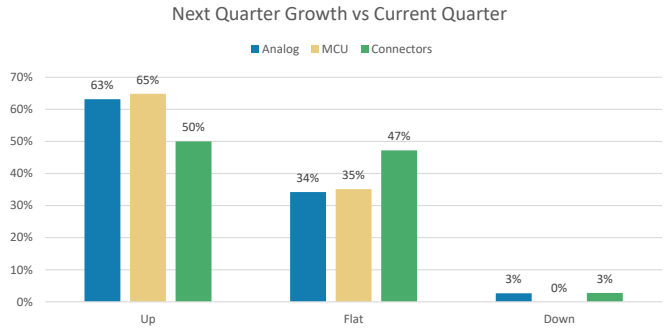
DecQ expectations remain healthy across Analog/MCU, even after accounting for seasonality

Exhibit 39: The majority of respondents anticipate results in-line with or stronger than seasonal...



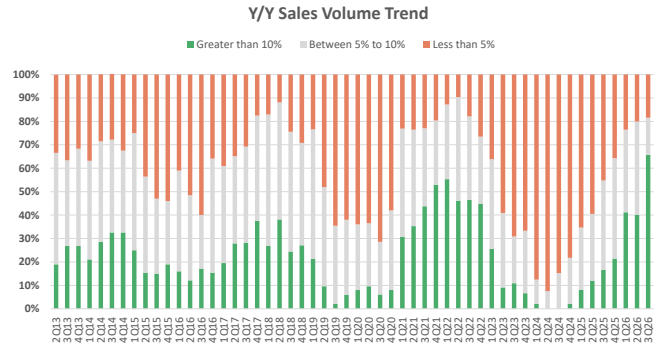
Source: AlphaWise, Morgan Stanley Research

Exhibit 40: ...while sequential DecQ growth expectations remain healthy



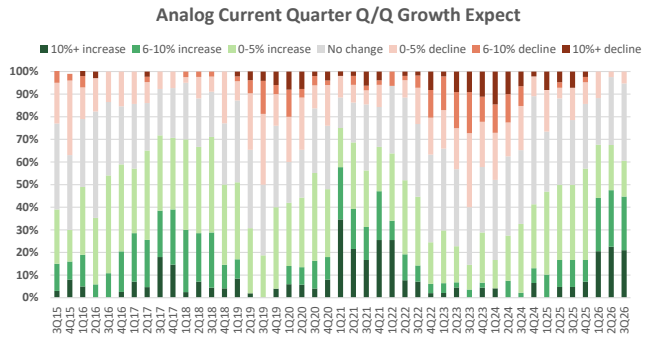
Source: AlphaWise, Morgan Stanley Research

Exhibit 41: A record 66% of respondents report y/y sales growth above 10%, up from 40% last quarter



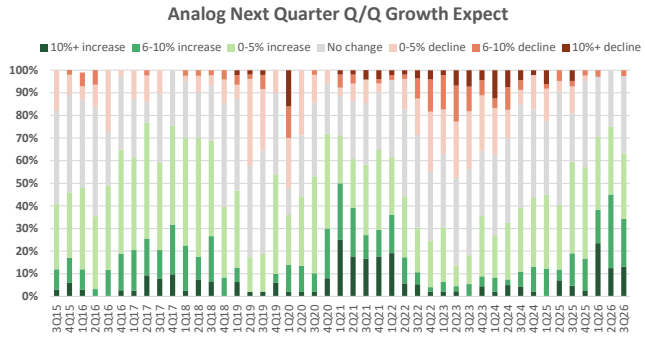
Source: AlphaWise, Morgan Stanley Research

Exhibit 42: Analog SepQ growth expectations remain healthy, with 95% anticipating flat-to-up sales…



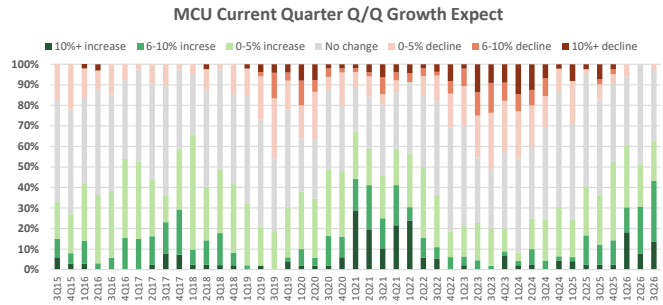
Source: AlphaWise, Morgan Stanley Research

Exhibit 43: …with 97% anticipating flat/up sales in DecQ



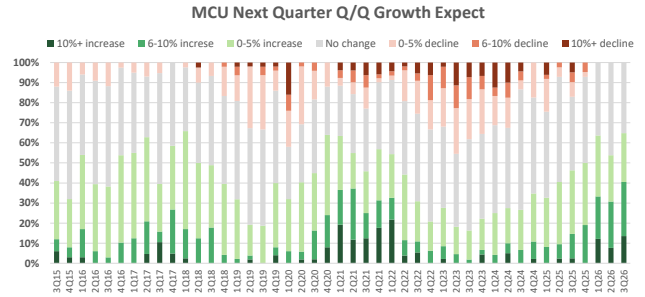
Source: AlphaWise, Morgan Stanley Research

Exhibit 44: MCU SepQ expectations strengthened, with 97% anticipating flat/up sales...



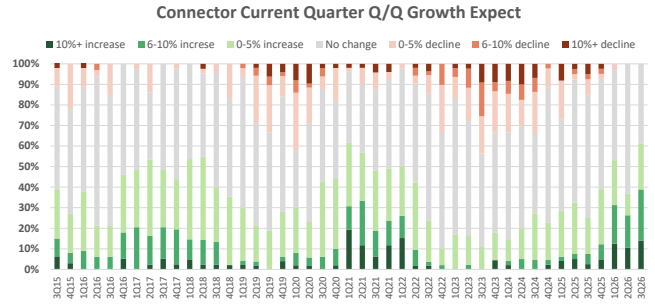
Source: AlphaWise, Morgan Stanley Research

Exhibit 45: …while DecQ growth expectations increased to 65% from 54%



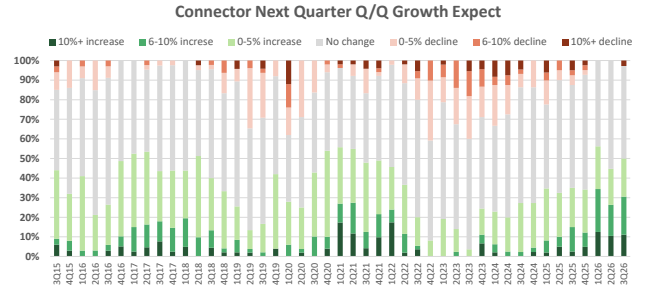
Source: AlphaWise, Morgan Stanley Research

Exhibit 46: Connector SepQ expectations remain constructive, with no respondents anticipating a decline...



Source: AlphaWise, Morgan Stanley Research

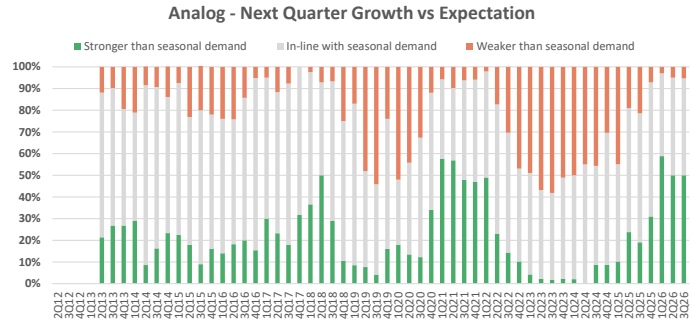
Exhibit 47: …while DecQ growth expectations strengthened to 50% from 45%



Source: AlphaWise, Morgan Stanley Research

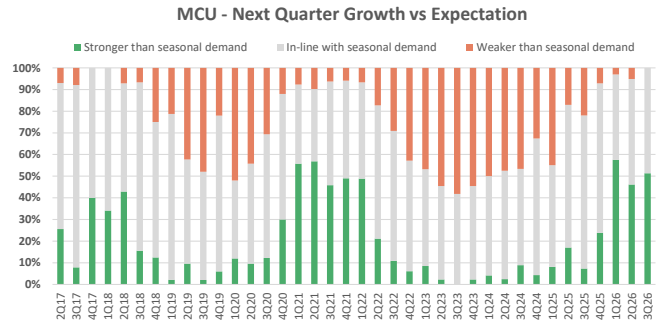
Above-seasonal expectations remain healthy across all product lines, with MCU & Connectors improving q/q

Exhibit 48: 50% of respondents expect a stronger-than-seasonal DecQ for Analog, unchanged q/q...



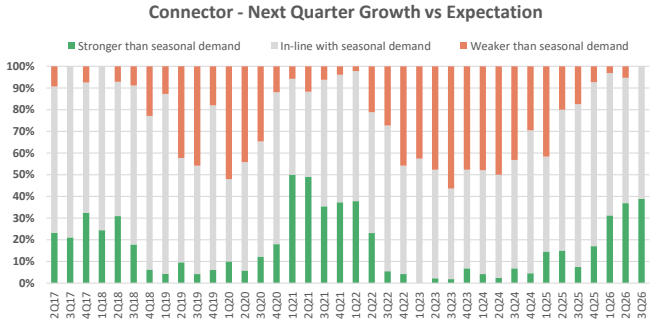
Source: AlphaWise, Morgan Stanley Research

Exhibit 49: ... while 51% of respondents expect a stronger-than-seasonal DecQ for MCU, up from 46% last quarter...



Source: AlphaWise, Morgan Stanley Research

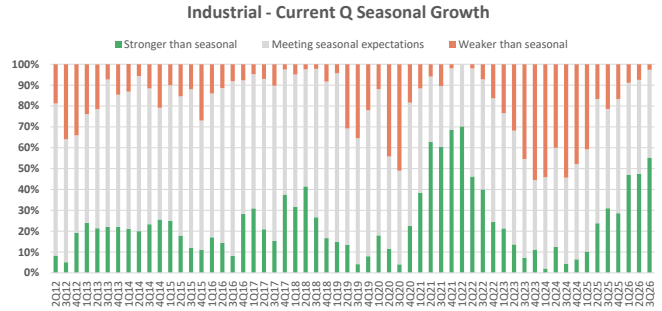
Exhibit 50: ... and 39% of respondents expect a stronger-than-seasonal DecQ for connectors, up from 37% last quarter



Source: AlphaWise, Morgan Stanley Research

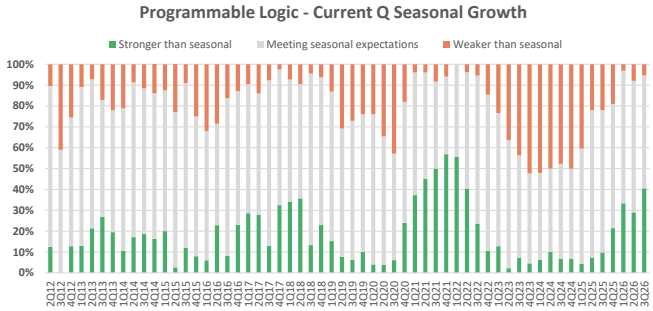
End-market trends improve, led by Industrial/PLD, while Communications/Auto remain largely in-line with seasonality

Exhibit 51: 55% expect stronger-than-seasonal Industrial sales, up from 48% last quarter...



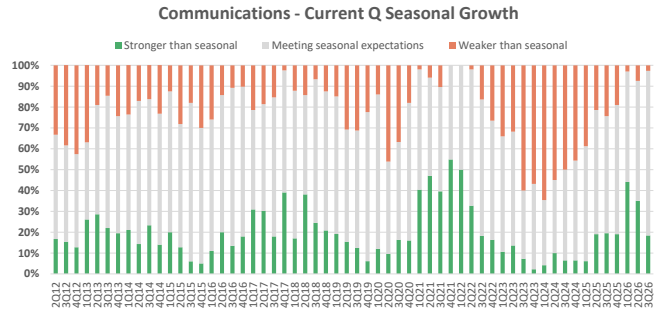
Source: AlphaWise, Morgan Stanley Research

Exhibit 52: ...with expectations for stronger-than-seasonal PLD sales increasing to 41% from 29%



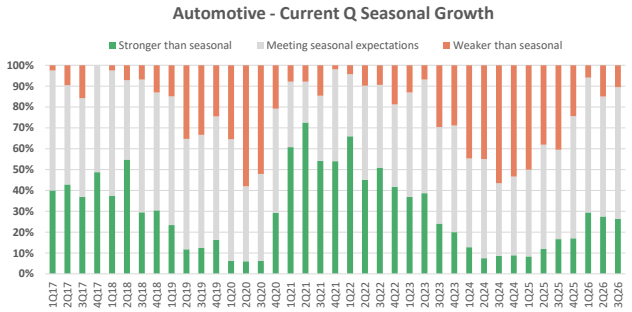
Source: AlphaWise, Morgan Stanley Research

Exhibit 53: Communications expectations moderate, with 79% now anticipating in-line seasonal demand vs. 58% last quarter...



Source: AlphaWise, Morgan Stanley Research

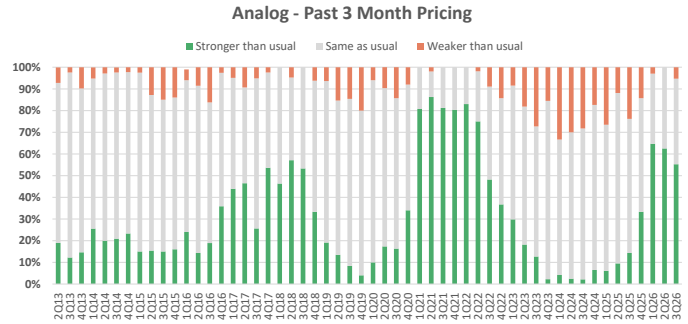
Exhibit 54: ...while expectations for Auto remain largely in-line with seasonality at 63% vs. 58% last quarter



Source: AlphaWise, Morgan Stanley Research

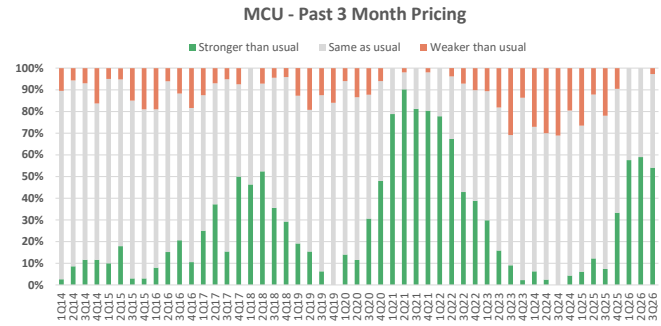
Pricing strength moderates, but remains elevated

Exhibit 55: Respondents seeing stronger-than-usual Analog pricing moderated to 55% from 63%, but remains elevated...



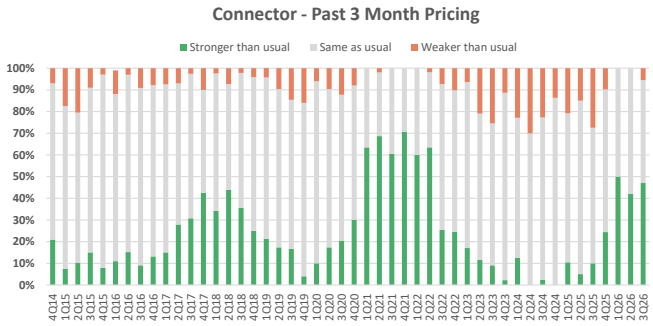
Source: AlphaWise, Morgan Stanley Research

Exhibit 56: ...similarly, MCU pricing moderated only modestly to 54% from 59%, but remains strong...



Source: AlphaWise, Morgan Stanley Research

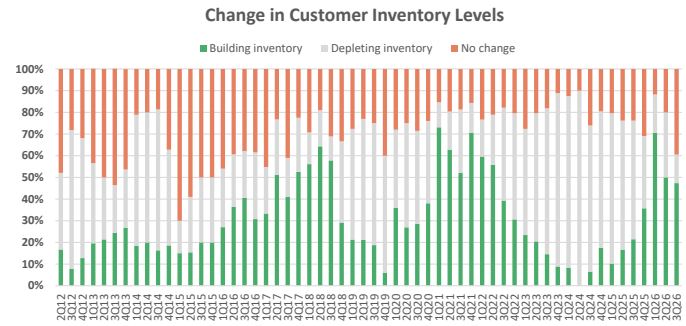
Exhibit 57: ...while Connector pricing remains largely stable despite some emerging price relief



Source: AlphaWise, Morgan Stanley Research

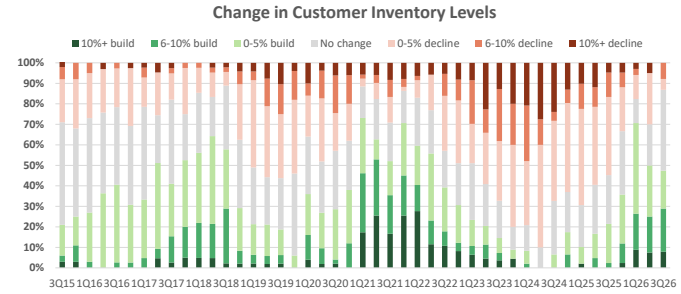
Across product lines, inventory depletion declined sharply, while building intentions remained healthy

Exhibit 58: Respondents looking to deplete inventory fell sharply to 13% from 30% last quarter...



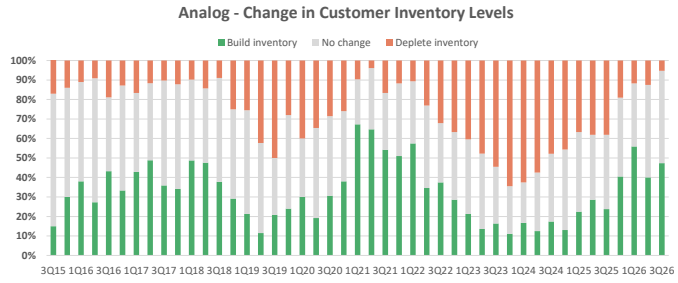
Source: AlphaWise, Morgan Stanley Research

Exhibit 59: ...with 8% of respondents looking to deplete by greater than 5%, up slightly from 5% last quarter



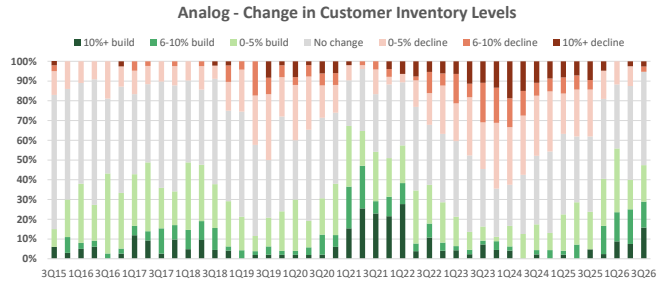
Source: AlphaWise, Morgan Stanley Research

Exhibit 60: In analog, 5% of respondents are looking to deplete inventory, down from 13% last quarter...



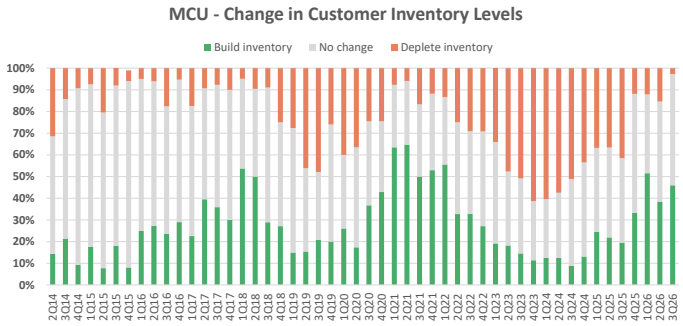
Source: AlphaWise, Morgan Stanley Research

Exhibit 61: …and 29% are looking to build inventory by more than 5%, up slightly from 25%



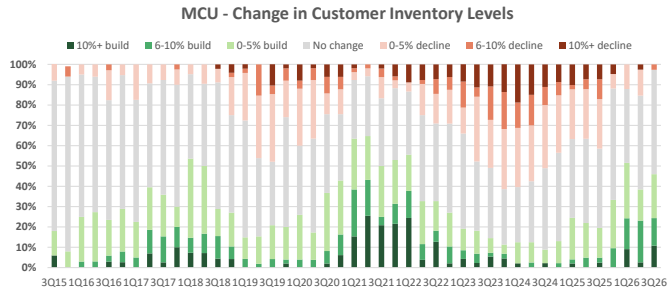
Source: AlphaWise, Morgan Stanley Research

Exhibit 62: In MCUs, 46% of respondents are looking to build inventory, up from 38% last quarter…



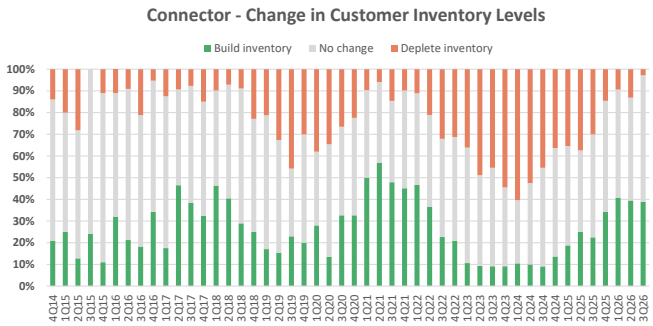
Source: AlphaWise, Morgan Stanley Research

Exhibit 63: …and 24% are looking to build by more than 5%, ~flat from 23%



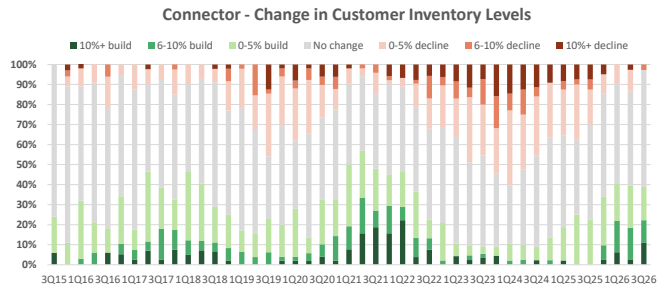
Source: AlphaWise, Morgan Stanley Research

Exhibit 64: In connectors, 3% of respondents are looking to deplete inventory, down from 13% last quarter…



Source: AlphaWise, Morgan Stanley Research

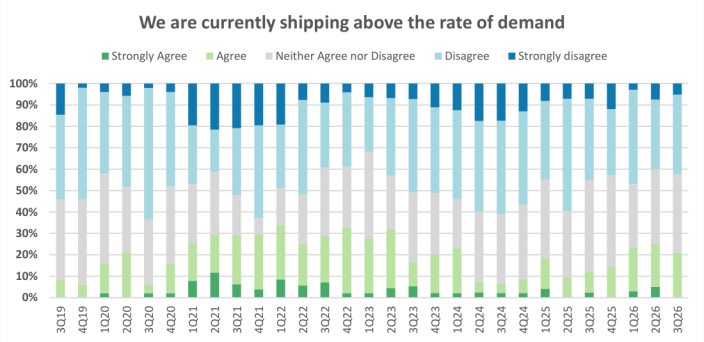
Exhibit 65: … and 22% are looking to build by more than 5%, up from 18%



Source: AlphaWise, Morgan Stanley Research

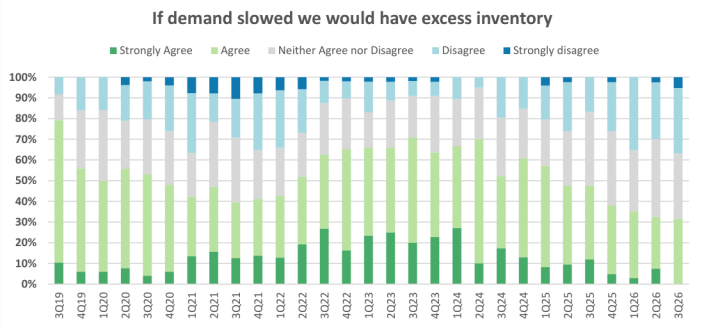
Inventory concerns remain contained, despite a modest increase in excess inventory risk

Exhibit 66: 21% of respondents believe they are shipping above the rate of demand, ~flat from 20%...



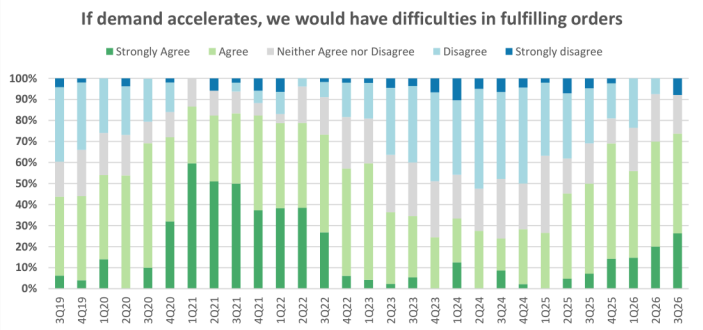
Source: AlphaWise, Morgan Stanley Research

Exhibit 67: ...while 32% of respondents believe that if demand slowed they would be holding excess inventory, up from 25% last quarter



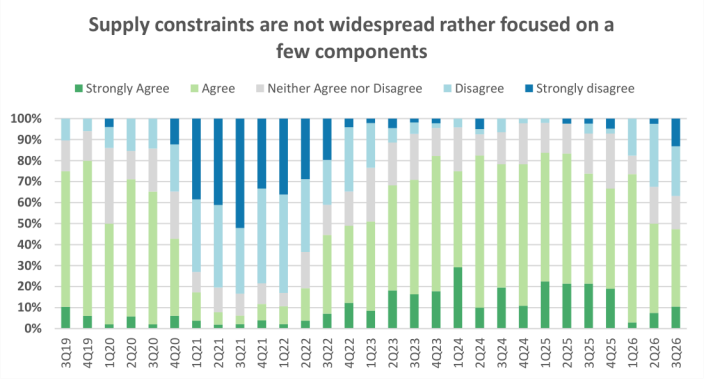
Source: AlphaWise, Morgan Stanley Research

Exhibit 68: 47% of respondents believe that they would have difficulty fulfilling orders if demand accelerates, ~in-line with 50% last quarter



Source: AlphaWise, Morgan Stanley Research

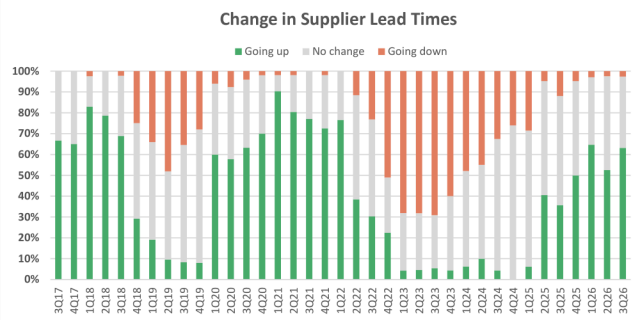
Exhibit 69: 63% of respondents believe that supply constraints are widespread, up from 57% last quarter



Source: AlphaWise, Morgan Stanley Research

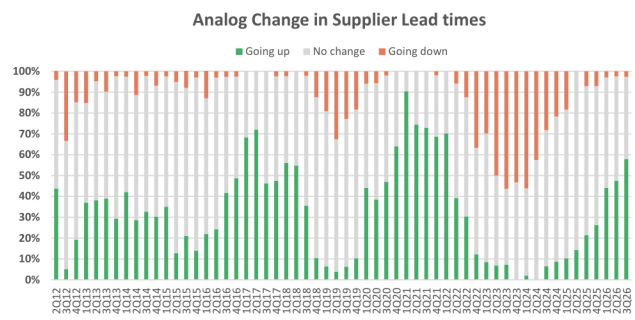
Lead time increases accelerated across product lines, led by MCU

Exhibit 70: 63% of respondents reported that supplier lead times are going up, up from 53% last quarter



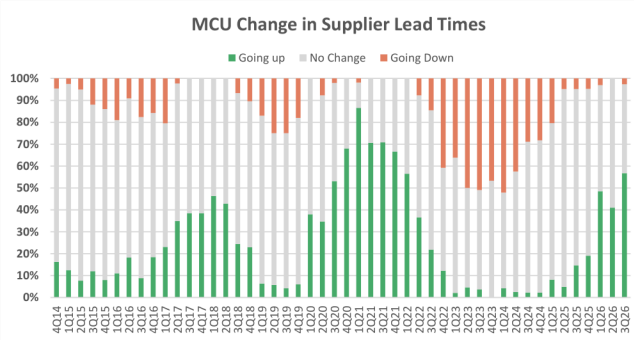
Source: AlphaWise, Morgan Stanley Research

Exhibit 71: 58% of respondents reported Analog lead times increasing, up from 48% last quarter...



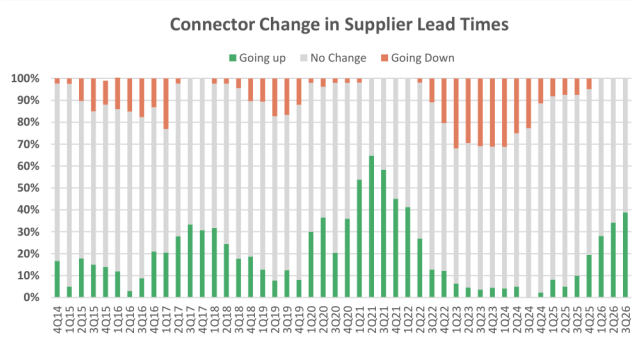
Source: AlphaWise, Morgan Stanley Research

Exhibit 72: ...with MCU seeing a similar trend as 57% of respondents reported increasing lead times, up from 41%...



Source: AlphaWise, Morgan Stanley Research

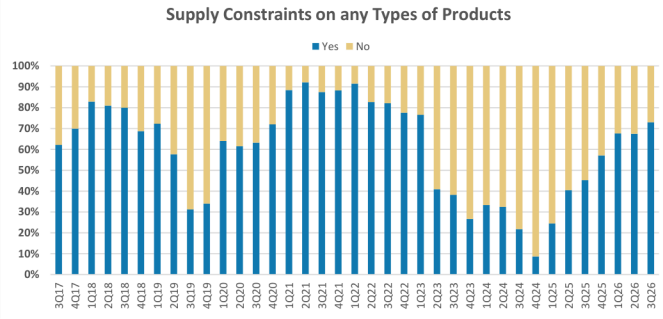
Exhibit 73: ...while connectors ticked up slightly to 39% reporting increasing lead times vs 34% last quarter



Source: AlphaWise, Morgan Stanley Research

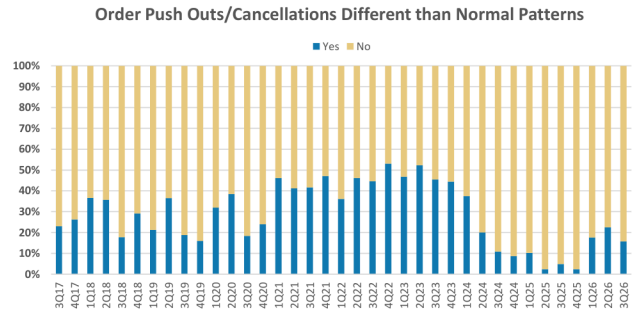
Supply constraints broaden across product lines

Exhibit 74: Respondents reporting supply constraints on at least one product increased to 73% from 68% last quarter...



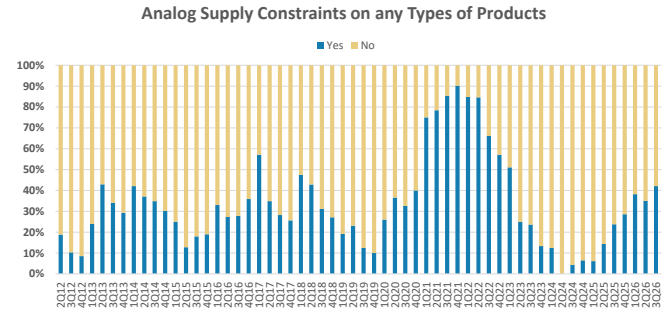
Source: AlphaWise, Morgan Stanley Research

Exhibit 75: ...while abnormal order push-outs/cancellations eased slightly to 16% from 23% last quarter



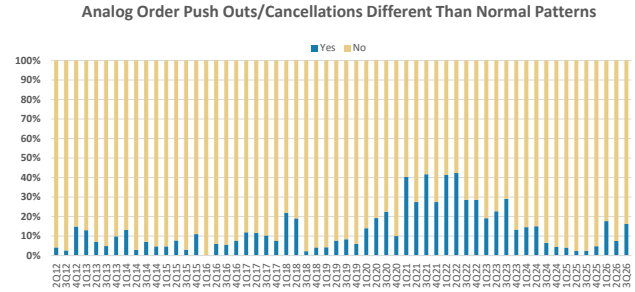
Source: AlphaWise, Morgan Stanley Research

Exhibit 76: Analog supply constraints increased to 42% from 35%...



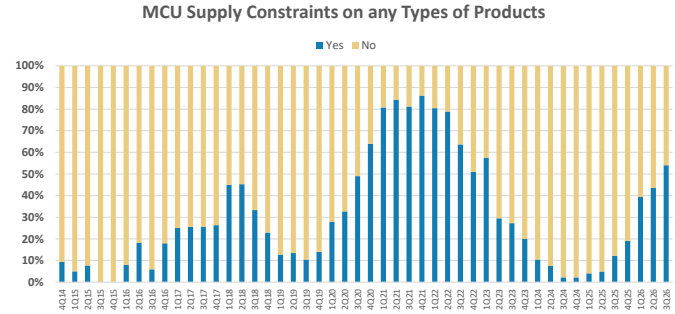
Source: AlphaWise, Morgan Stanley Research

Exhibit 77: ...with push-outs and cancellations increasing to 16% from 8%



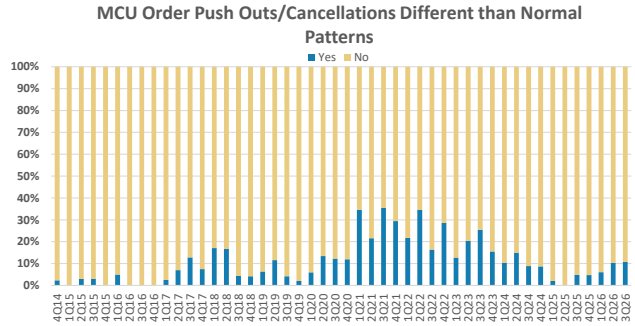
Source: AlphaWise, Morgan Stanley Research

Exhibit 78: MCU supply constraints increased to 54% from 44%...



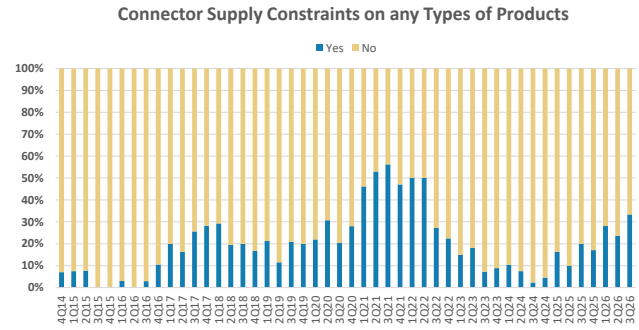
Source: AlphaWise, Morgan Stanley Research

Exhibit 79: ...while MCU push-outs/cancellations remained ~flat at 11% vs. 10% last quarter



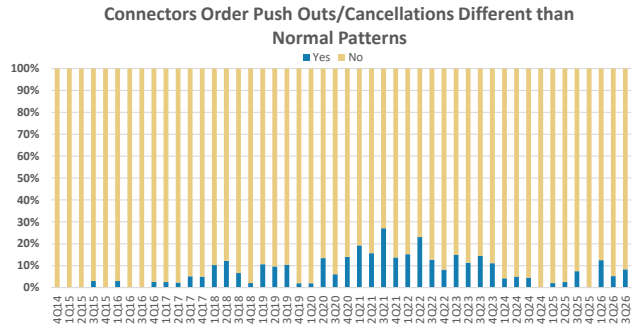
Source: AlphaWise, Morgan Stanley Research

Exhibit 80: Connector supply constraints increased to 33% from 24%...



Source: AlphaWise, Morgan Stanley Research

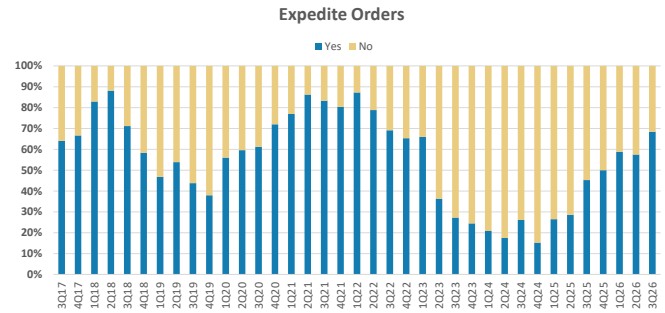
Exhibit 81: ...with Connector push-outs/cancellations ticking up slightly to 8% from 5%



Source: AlphaWise, Morgan Stanley Research

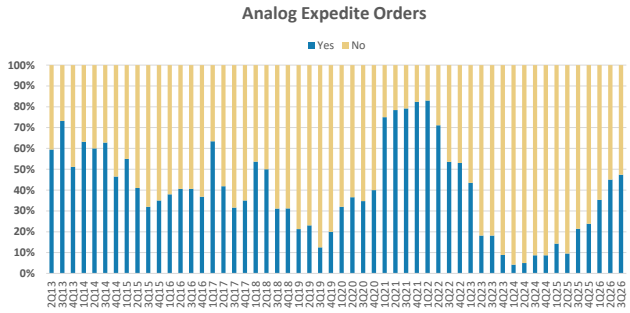
Expedites accelerate sharply, led by MCU, while double ordering eases slightly

Exhibit 82: Overall expedite orders increased to 68% from 58%



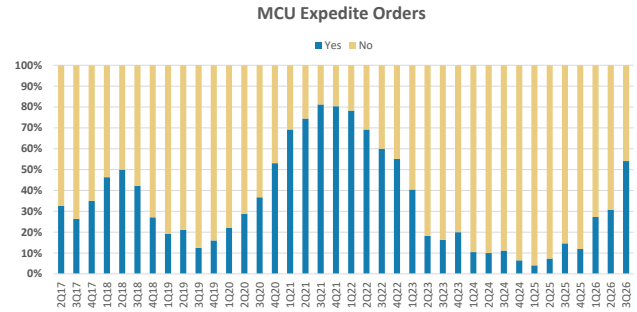
Source: AlphaWise, Morgan Stanley Research

Exhibit 83: Analog order expedites increased slightly to 47% from 45%



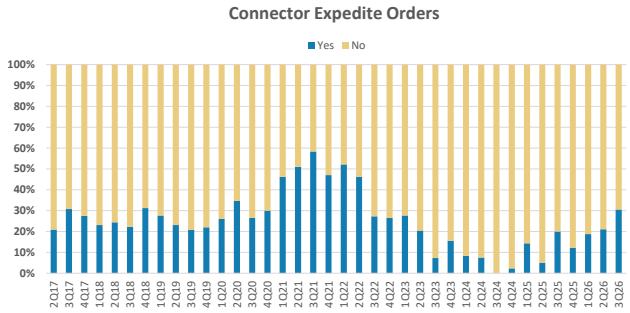
Source: AlphaWise, Morgan Stanley Research

Exhibit 84: MCU expedites jumped to 54% from 31% last quarter



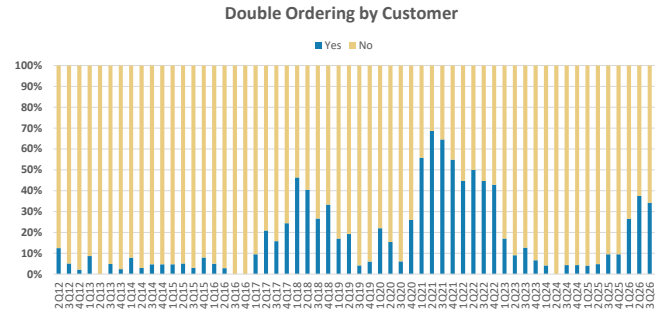
Source: AlphaWise, Morgan Stanley Research

Exhibit 85: Connector order expedites increased to 31% from 21%



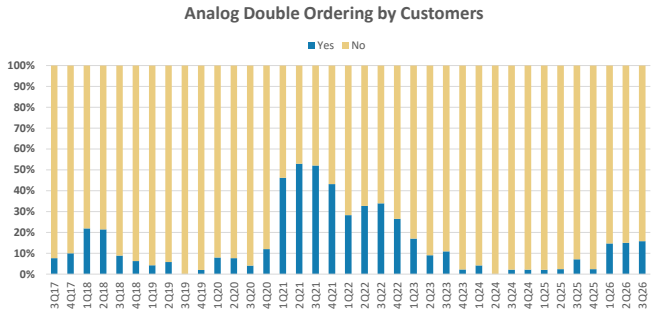
Source: AlphaWise, Morgan Stanley Research

Exhibit 86: Overall double ordering decreased slightly to 34% from 38%, though remains at elevated levels



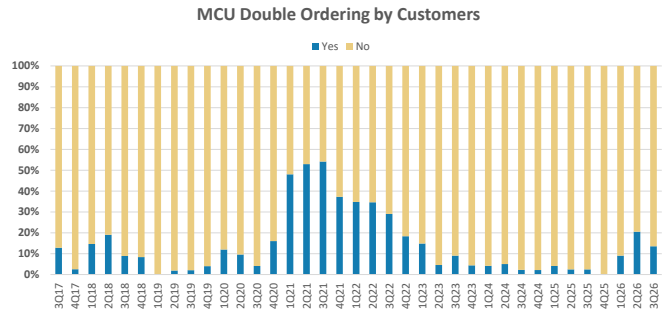
Source: AlphaWise, Morgan Stanley Research

Exhibit 87: Analog double ordering remained in-line with last quarter at 16%



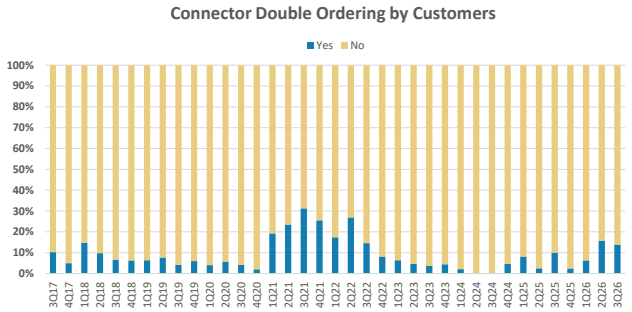
Source: AlphaWise, Morgan Stanley Research

Exhibit 88: MCU double ordering decreased to 14% from 21%



Source: AlphaWise, Morgan Stanley Research

Exhibit 89: Connector double ordering decreased marginally to 14% from 16%

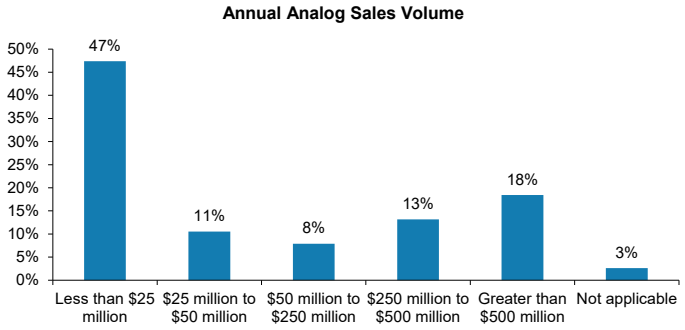


Source: AlphaWise, Morgan Stanley Research

Survey Sample Profile

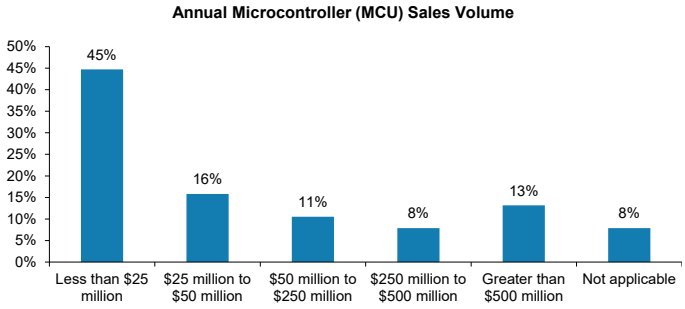
Morgan Stanley AlphaWise surveyed 38 respondents in the North American distribution channel to gauge market conditions. % of respondents with sales volume of over $250mn were as follows; Analog 32%, MCU 21%, PLD 16%, Discrete 13%, Connectors 18%.

Exhibit 90: Analog Survey Respondents



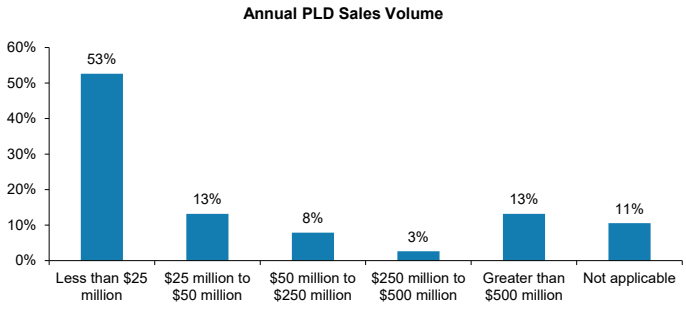
Source: AlphaWise, Morgan Stanley Research

Exhibit 91: MCU Survey Respondents



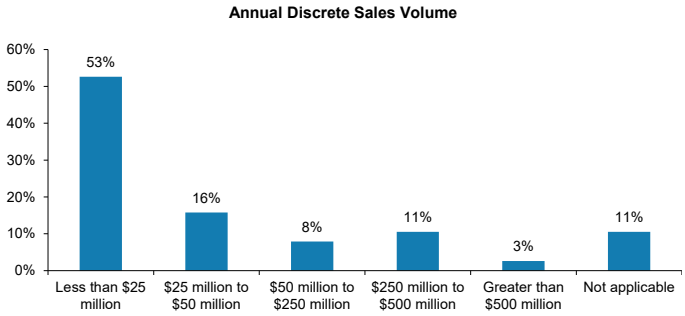
Source: AlphaWise, Morgan Stanley Research

Exhibit 92: PLD Survey Respondents



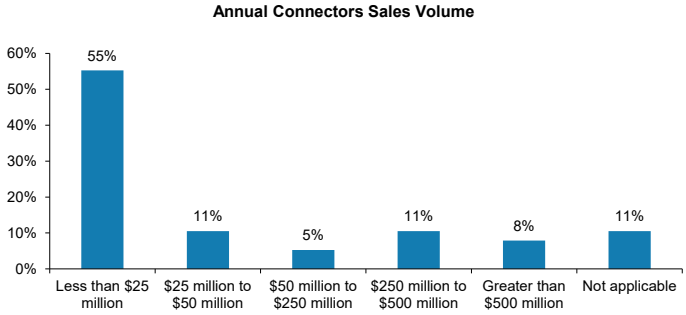
Source: AlphaWise, Morgan Stanley Research

Exhibit 93: Discrete Survey Respondents



Source: AlphaWise, Morgan Stanley Research

Exhibit 94: Connector Survey Respondents



Source: AlphaWise, Morgan Stanley Research

Valuation Methodology and Risks

Analog Devices Inc. (ADI.O)

We value ADI at 27x CY27e MW EPS of $16.95, a premium to analog peers given higher GM% and OPM%.

Risks to Upside

- Further margin improvement
- Semi content growth in EVs & share gain
- Industrial recovery stronger than anticipated

Risks to Downside

- Market share loss in Automotive and Industrial
- Increasing competition in auto semi
- Weakness in consumer exposed end-markets

Microchip Technology Inc. (MCHP.O)

Our PT of $103 represents an early-mid-cycle multiple of 25x on CY27e Non-GAAP EPS of $4.10.

Risks to Upside

- Continued share gains in MCU drive cross selling synergies with analog and FPGA
- Shareholder returns expand ahead of market expectation

Risks to Downside

- Broad based semi down cycle lasts longer than anticipated
- Debt and net leverage to increase
- Company loses market share to incremental China capacity

NXP Semiconductor NV (NXPI.O)

Our PT represents 18x our CY27 Non-GAAP EPS estimate, a slight premium to the company's historical average which we think is justified given the company's downcycle management and set up to outperform peers in 2026/2027.

Risks to Upside

- Increasing momentum in auto growth drivers (BMS, radar & high-end processors)
- Stronger than expected data center ramps from design wins
- Gross margin expands to the low-60s on the back of manufacturing and portfolio optimization

Risks to Downside

- Increased competition in auto slows growth
- Industrial & IoT downturn extends longer than anticipated
- Weakness in consumer end-markets

Texas Instruments (TXN.O)

Represents 26x our CY27e EPS of $9.80, reflecting mid-cycle positioning, a slight premium to its long-term average of 24x and ~in-line with large-cap analog peers.

Risks to Upside

- Increasing domestic capacity leads to outsized market share gains.
- Recovery in consumer markets (personal electronics and embedded).
- Continued content gains in Auto and Industrial end-markets.

Risks to Downside

- Continued market share loss in the embedded business.
- Pricing and market share pressure from China local capacity.
- Market share loss to ADI in the analog business.

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The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Shane Brett; Nicole Kozhukhov; Joseph Moore; Ella Tulchinsky; Mason Wayne.

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(as of September 30, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

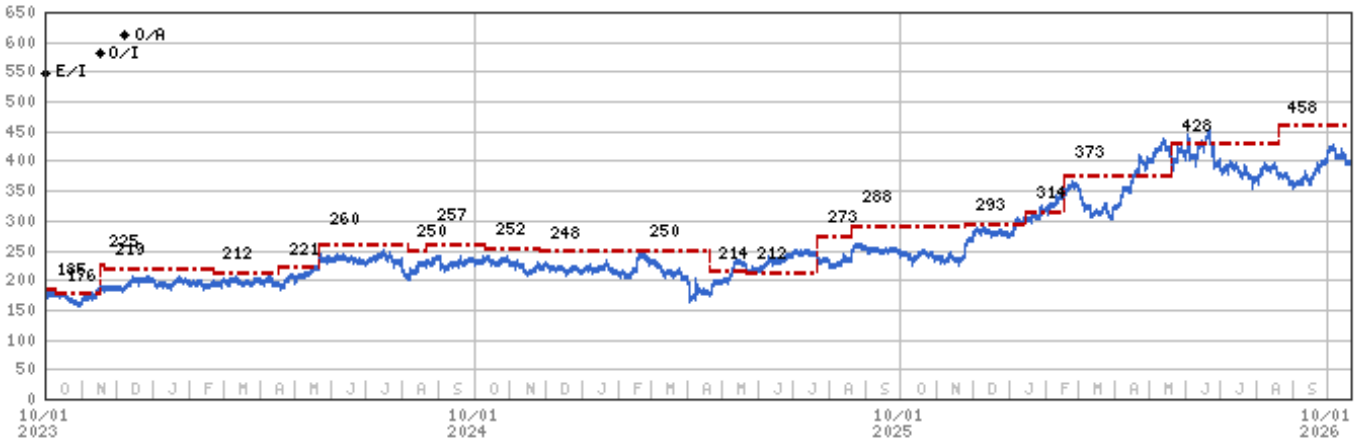
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Analog Devices Inc. (ADI.O) - As of 10/20/26 GMT in USD
Industry : Semiconductors



Stock Rating History: 10/1/21 : E/I; 11/16/23 : O/I; 12/7/23 : O/A

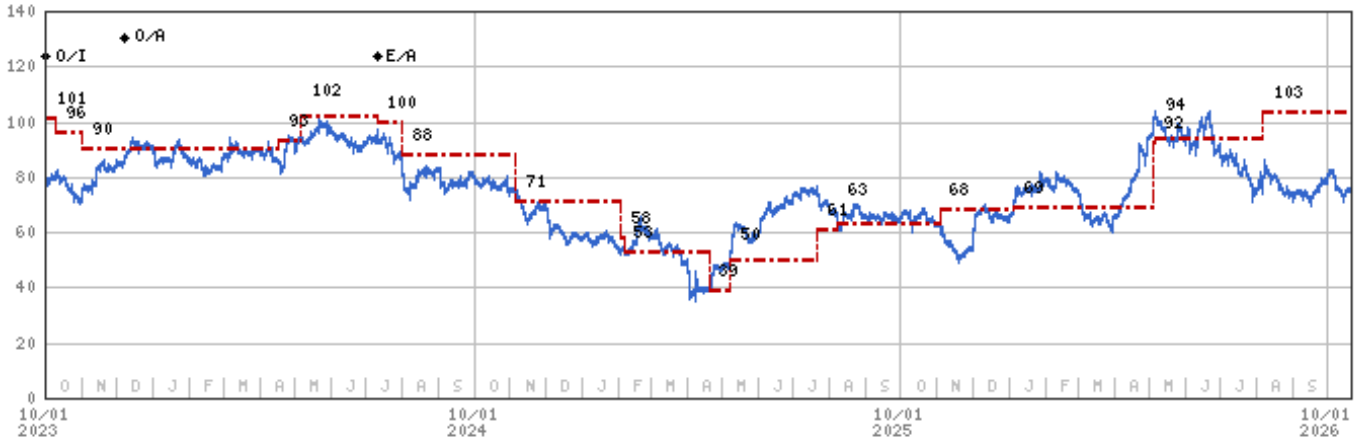
Price Target History: 9/13/21 : 179; 11/23/21 : 194; 2/16/22 : 196; 5/15/22 : 178; 5/18/22 : 186; 6/9/22 : 173; 11/22/22 : 177; 2/15/23 : 210; 5/24/23 : 195; 8/23/23 : 185; 10/10/23 : 176; 11/16/23 : 225; 11/21/23 : 219; 2/21/24 : 212; 4/17/24 : 221; 5/22/24 : 260; 8/5/24 : 250; 8/21/24 : 257; 10/10/24 : 252; 11/26/24 : 248; 2/19/25 : 250; 4/20/25 : 214; 5/22/25 : 212; 7/21/25 : 273; 8/20/25 : 288; 11/25/25 : 293; 1/16/26 : 314; 2/18/26 : 373; 5/20/26 : 428; 8/19/26 : 458

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Microchip Technology Inc. (MCHP.O) - As of 10/20/26 GMT in USD
Industry : Semiconductors



Stock Rating History: 10/1/21 : O/I; 12/7/23 : O/A; 7/10/24 : E/A

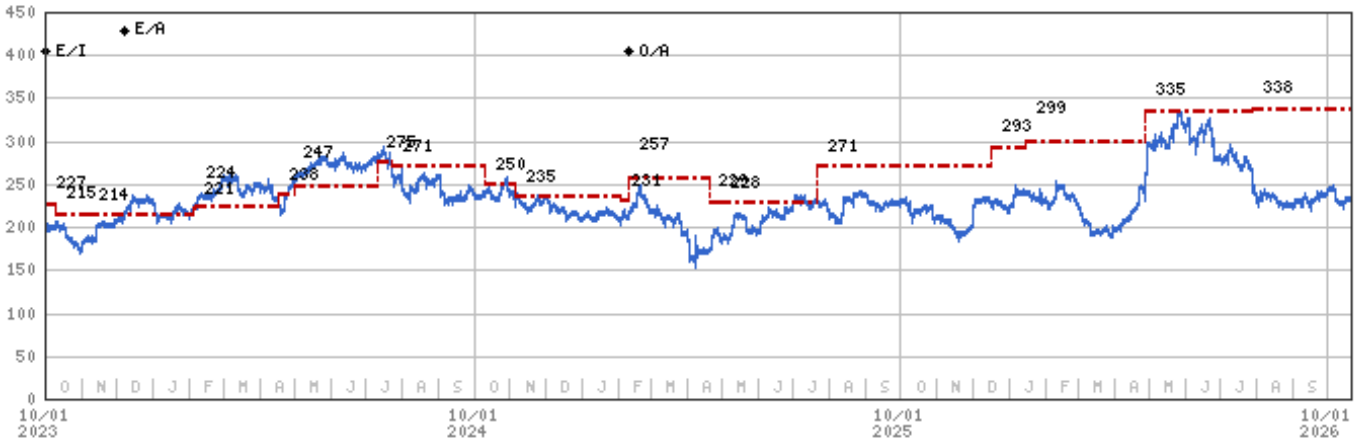
Price Target History: 8/4/21 : 95; 11/4/21 : 100; 5/8/22 : 85; 6/9/22 : 81; 8/2/22 : 82; 11/3/22 : 80; 2/2/23 : 99; 5/4/23 : 97; 8/3/23 : 101; 10/10/23 : 96; 11/2/23 : 90; 4/17/24 : 93; 5/6/24 : 102; 7/10/24 : 100; 8/1/24 : 88; 11/5/24 : 71; 2/3/25 : 58; 2/6/25 : 53; 4/20/25 : 39; 5/8/25 : 50; 7/21/25 : 61; 8/8/25 : 63; 11/3/25 : 68; 1/5/26 : 69; 5/4/26 : 92; 5/7/26 : 94; 8/6/26 : 103

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

NXP Semiconductor NV (NXPI.O) - As of 10/20/26 GMT in USD
Industry : Semiconductors



Stock Rating History: 10/1/21 : E/I; 12/7/23 : E/A; 2/11/25 : O/A

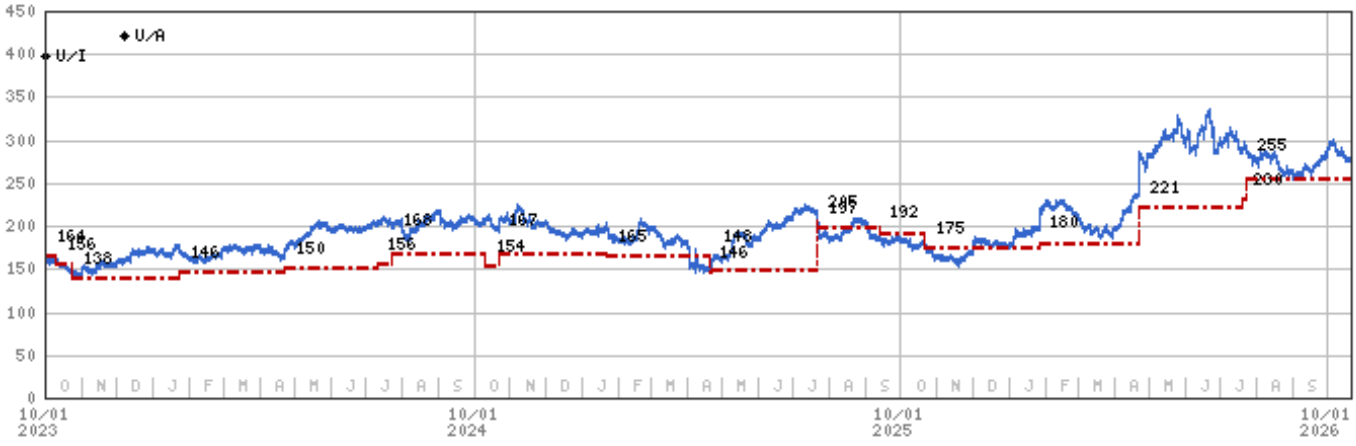
Price Target History: 8/4/21 : 232; 11/2/21 : 241; 12/14/21 : 256; 2/1/22 : 262; 5/1/22 : 188; 5/3/22 : 194; 6/9/22 : 180; 7/26/22 : 183; 11/1/22 : 174; 1/31/23 : 177; 5/2/23 : 180; 7/25/23 : 227; 10/10/23 : 215; 11/7/23 : 214; 2/4/24 : 221; 2/6/24 : 224; 4/17/24 : 238; 4/30/24 : 247; 7/10/24 : 275; 7/23/24 : 271; 10/10/24 : 250; 11/5/24 : 235; 2/4/25 : 231; 2/11/25 : 257; 4/20/25 : 229; 4/29/25 : 228; 7/21/25 : 271; 12/17/25 : 293; 1/16/26 : 299; 4/28/26 : 335; 7/29/26 : 338

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

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Texas Instruments (TXN.O) - As of 10/20/26 GMT in USD
Industry : Semiconductors



Stock Rating History: 10/1/21 : U/I; 12/7/23 : U/A

Price Target History: 7/22/21 : 193; 10/26/21 : 195; 1/25/22 : 198; 4/25/22 : 181; 4/26/22 : 170; 6/9/22 : 155; 7/26/22 : 160; 10/25/22 : 152; 1/24/23 : 166; 4/25/23 : 159; 7/25/23 : 164; 10/10/23 : 156; 10/24/23 : 138; 1/23/24 : 146; 4/23/24 : 150; 7/10/24 : 156; 7/23/24 : 168; 10/10/24 : 154; 10/22/24 : 167; 1/23/25 : 165; 4/20/25 : 146; 4/23/25 : 148; 7/21/25 : 205; 7/22/25 : 197; 9/12/25 : 192; 10/21/25 : 175; 1/27/26 : 180; 4/22/26 : 221; 7/20/26 : 230; 7/23/26 : 255

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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INDUSTRY COVERAGE: Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/20/2026)
Ella Tulchinsky		
Infleqtion, Inc. (INFQ.N)	E (10/15/2026)	$8.66
Xanadu Quantum Technologies Limited (XNDU.O)	E (10/15/2026)	$2.85
Joseph Moore		
Advanced Micro Devices (AMD.O)	E (06/09/2024)	$534.74
Aeva Technologies Inc (AEVA.O)	E (07/19/2021)	$10.27
Allegro Microsystems Inc (ALGM.O)	O (02/13/2026)	$33.91
Ambarella Inc (AMBA.O)	O (03/29/2016)	$63.22
Amkor Technology Inc (AMKR.O)	E (11/08/2023)	$41.70
Analog Devices Inc. (ADI.O)	O (11/16/2023)	$397.05
Astera Labs Inc (ALAB.O)	O (05/11/2025)	$284.80
Broadcom Inc. (AVGO.O)	O (06/09/2024)	$334.37
Cerebras Systems (CBRS.O)	O (06/08/2026)	$118.24
GlobalFoundries Inc (GFS.O)	E (10/28/2024)	$47.01
Intel Corporation (INTC.O)	E (02/22/2023)	$91.69
IonQ Inc (IONQ.N)	E (04/25/2023)	$31.26
Marvell Technology Group Ltd (MRVL.O)	E (09/14/2015)	$240.42
Microchip Technology Inc. (MCHP.O)	E (07/10/2024)	$75.85
Micron Technology Inc. (MU.O)	O (10/06/2025)	$896.07
Navitas Semiconductor Corp (NVTS.O)	U (04/06/2025)	$8.74
NVIDIA Corp. (NVDA.O)	O (03/16/2023)	$216.47
NXP Semiconductor NV (NXPI.O)	O (02/11/2025)	$237.60
ON Semiconductor Corp. (ON.O)	++	$75.62
Qualcomm Inc. (QCOM.O)	E (06/24/2026)	$175.71
Quantinuum (QNT.O)	O (10/15/2026)	$35.68
SanDisk Corporation. (SNDK.O)	O (03/03/2025)	$1,365.38
Semtech Corp. (SMTC.O)	E (04/06/2025)	$136.20
Silicon Laboratories Inc. (SLAB.O)	E (01/19/2021)	$221.10
SiTime Corporation (SITM.O)	O (09/16/2026)	$497.11
Skyworks Solutions Inc (SWKS.O)	E (11/28/2018)	$81.30
Texas Instruments (TXN.O)	U (04/13/2020)	$277.78
Wolfspeed, INC (WOLF.N)	NR (04/06/2025)	$23.50
Lee Simpson		
Arm Holdings plc (ARM.O)	E (04/07/2026)	$221.06
Cadence Design Systems Inc (CDNS.O)	O (02/14/2024)	$319.87
PDF Solutions Inc (PDFS.O)	O (09/08/2026)	$41.18
Synopsys Inc. (SNPS.O)	O (09/08/2026)	$447.62

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